

Questionnaire design

- Part 1 -

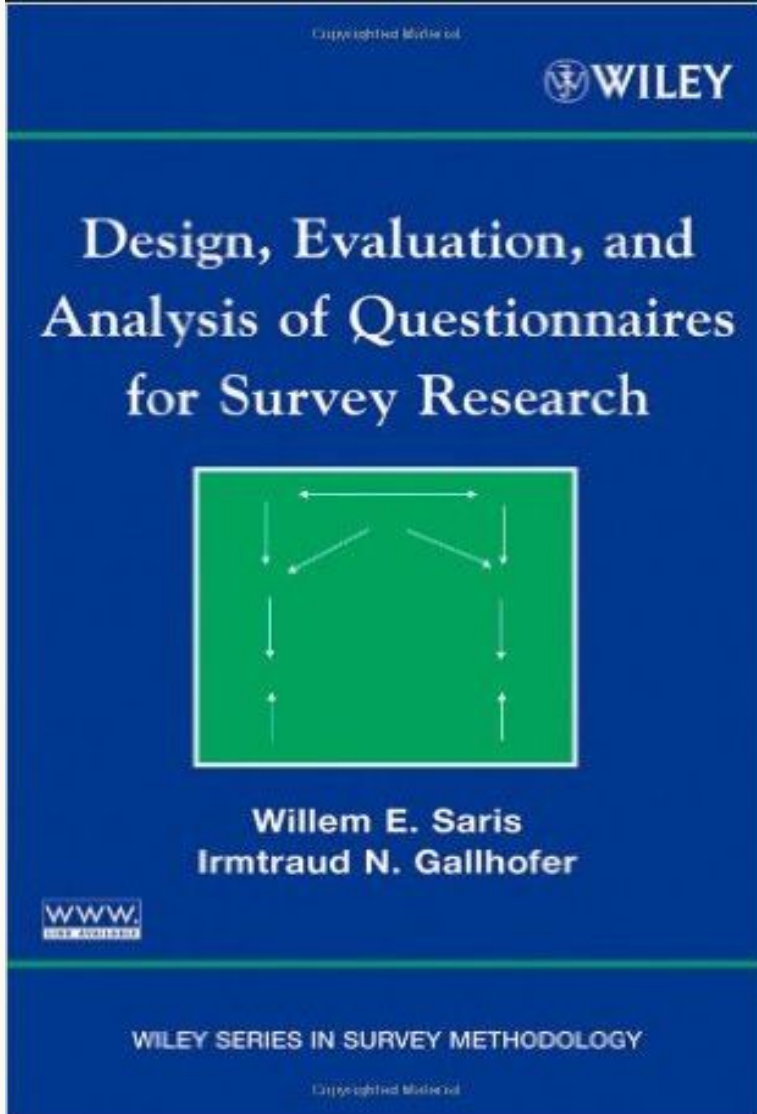
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11 May 2026

Aims

- Demonstrate the importance of good question & the consequences of bad questions
- Share key principles and empirical evidence that can guide you to create best questionnaires
- Introduce you to a procedure that should guarantee that you measure what you are interested in / a systematic way of formulating survey items
- Show you all the choices you have and decision you have to take

Main reference



Revilla, M., D. Zavala-Rojas, and W. Saris (2016). “Creating a good question: How to use cumulative experience”. In C. Wolf, D. Joye, T.W. Smith and Y.-C. Fu (editors), *The SAGE-Handbook of Survey Methodology*. SAGE. ISBN: 9781446282663. Chapter 17, pp.236-254.

Weber, W., I. Gallhofer, and W. Saris (2020) “Design of Survey Questions” SAGE Research Methods Foundations, 2020. SAGE Research Methods.

Present situation

- nowadays we are confronted with all kinds of surveys
- some parts of the world are considered “oversurveyed”

Present situation: surveys everywhere

COVID-19 Public Opinion Research | YouGov

https://wapor.org/resources/covid-19-public-opinion-research/

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COVID-19 Public Monitor | YouGov

today.yougov.com/covid-19

YouGov

All categories

COVID-19 Public Monitor

As a global public opinion organisation, YouGov is privileged that so many people around the world share their views and behaviours with us every day. At this time we're putting that full focus on gathering [information about COVID-19](#), asking people to [share their experiences](#) of the global pandemic and using that unique insight to [provide health organisations with data](#) that helps them understand and fight the spread of the virus. Data from the *COVID-19 behavior tracker* by Imperial College London and YouGov is accessible on [GitHub](#).

TRACKER

COVID-19 fears



TRACKER

COVID-19: government handling and confidence in health authorities

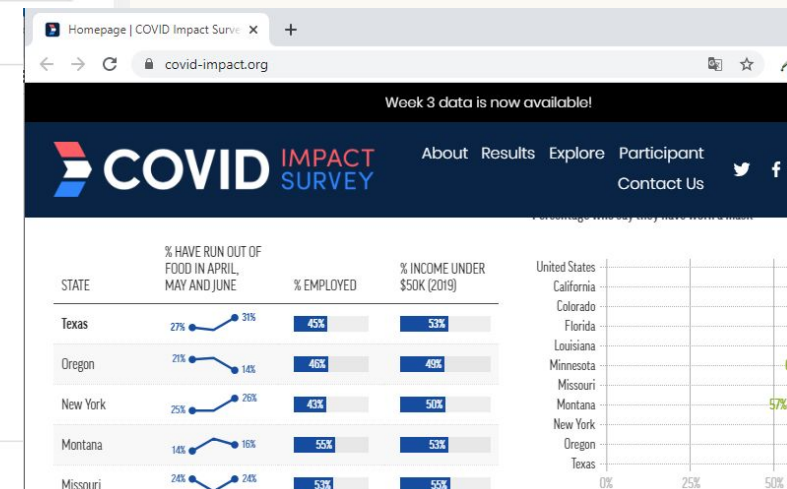


DATA

COVID-19 behavioural data for health organisations, media and

TRACKER

COVID-19: Support for actions governments could take



Present situation: with varying degrees of quality

Natalie Jackson on Twitter: "Re: ..."

twitter.com/nataliemj10/status/1273004487079342081/photo/1

In general, what is your level of concern related to the current coronavirus outbreak?

☐ Extremely concerned

☐ Moderately concerned

☒ Neutral

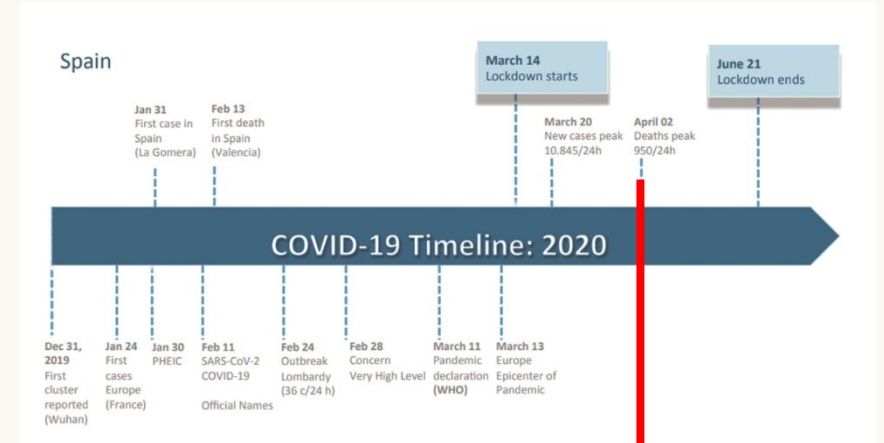
☐ Slightly concerned

☐ Not at all concerned

twitter.com/nataliemj10/status/1273004487079342081/photo/3

How are you feeling about the coronavirus?

Concerned	☐	☒	Hopeful
Confused	☐	☒	Clear
Uneasy	☐	☒	Composed
Upset	☐	☒	Calm
Insecure	☐	☒	Safe
Vulnerable	☐	☒	Protected
Scared	☐	☒	Peaceful



YouGov

g4-emea.yougov.com/v2ZT...

¿Qué probabilidades hay de que en los próximos 30 días usted...?

compre un producto para el cuidado de la piel

Muy probable Probable

Algo probable No muy probable

Nada probable

YouGov

g4-emea.yougov.com/v2ZT...

¿Qué probabilidades hay de que en los próximos 30 días usted...?

compre un coche

Muy probable Probable

Algo probable No muy probable

Nada probable

YouGov

g4-emea.yougov.com/v2ZT...

What are the chances that in the next 30 days you will... eat at a fast food restaurant

Very likely Likely

Somewhat likely Not very likely

Not at all likely

>

ENG 12:41 02/04/2020

Present situation

→ here we look at the complexity of the decisions which have to be made to do proper surveys with the focus on questionnaire design

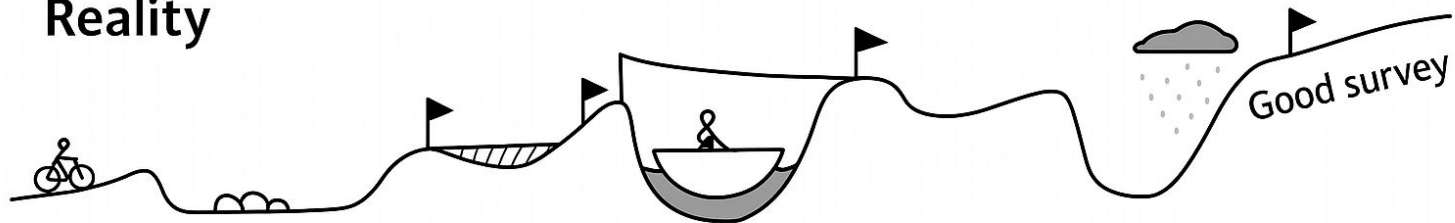
How people imagine questionnaire design



How people imagine questionnaire design with AI



Reality



Why is questionnaire design so important?

- Because the **quality** and **comparability** of the data, and thus the **conclusion** we draw based on this data, vary depending on:
 - The exact formulation of the questions
 - The presentation of the questions
 - The translation of the questions
 - The order of the questions
 - Etc.

What's a survey?

- A survey is data collection operation that gathers information from human respondents by means of a standardized questionnaire
- in which the interest is in characteristics of a population rather than particular individuals
- while the results are based on a sample of the population rather than the population itself.

Presser (1984: 95)

An illustration

British European Social Survey (ESS) pilot study

Concept: satisfaction

Economy	<i>On the whole, how satisfied are you with the present state of the economy in GB?</i>
Government	<i>Now think about the national government. How satisfied are you with the way it is doing its job?</i>
Democracy	<i>And on the whole, how satisfied are you with the way democracy works in GB?</i>

Three alternative response scales

Method 1 (M_1):

Very satisfied

Fairly satisfied

Fairly dissatisfied

Very dissatisfied

Method 2 (M_2):

very dissatisfied

0

1

2

3

4

5

6

7

8

9

very satisfied

10

Method 3 (M_3):

Not at all satisfied

Satisfied

Rather satisfied

Very satisfied

Does it matter what we choose?

- The combination of 3 questions (traits) and 3 response scales (methods) gives 9 different survey items
- All items were presented to a British sample (485 people)
 - Differences in responses **cannot** be due to sampling fluctuation (same respondents)
 - But only to formulation of the items

It matters for the distributions

Method 1 (M₁):
Very satisfied
Fairly satisfied
Fairly dissatisfied
Very dissatisfied

Method 2 (M₂):



Method 3 (M₃):
Not at all satisfied
Satisfied
Rather satisfied
Very satisfied

	Economy			Government		
	M1	M2	M3	M1	M2	M3
Dissatisfied	167	134	99	268	208	169
Neutral	-	102	-	-	100	-
Satisfied	273	193	320	191	128	258
Very Satisfied	26	7	12	11	2	4
Missing	19	49	54	15	47	54

No neutral but +100

More than twice higher

It matters for the correlations

Eco GovDem

Method 1

Eco	1.0		
Gov	.481	1.0	
Dem	.373	.552	1.0

Method 2

Eco	1.0		
Gov	.642	1.0	
Dem	.612	.693	1.0

Method 3

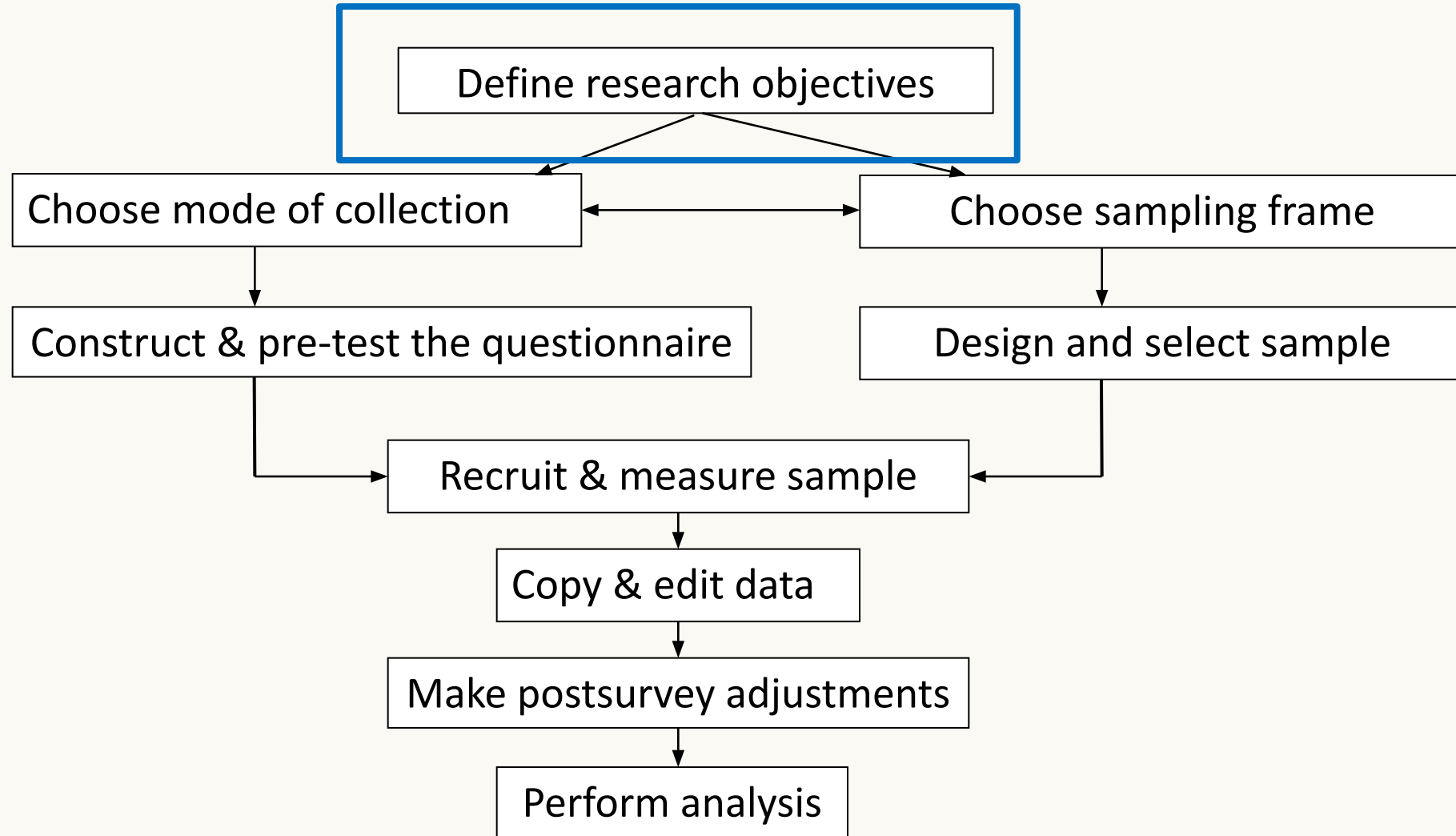
Eco	1.0		
Gov	.556	1.0	
Dem	.514	.558	1.0

→ Method matters!

→ Which one is the best?

STEPS OF SURVEY DESIGN

A survey from a process perspective



Research objectives

- **What** are you interested in finding out?
 - **Whom** do you want to study?
 - **Where** are these people or organizations located?
 - **When** do you want to do the survey?
 - **What** do you expect to learn and why?
- **Research question(s)!**
- **Hypotheses or expectations**

Research objectives

- Surveys can be used for many topics
- An important choice is whether one would like to do a *descriptive* or *explanatory* study

Descriptive or analytical?

Descriptive	Analytical
Focuses on <i>what?</i>	Focuses on <i>why?</i>
Is more exploratory	Is more explanatory
Profile characteristics of group	Analyze why group has characteristics
Assumes no statistical hypothesis	Assumes a statistical hypothesis
Does not require comparisons (between groups or over time)	Requires comparisons (between groups or over time)
<i>What do people think of X?</i>	<i>Why are people in favor or against X?</i>

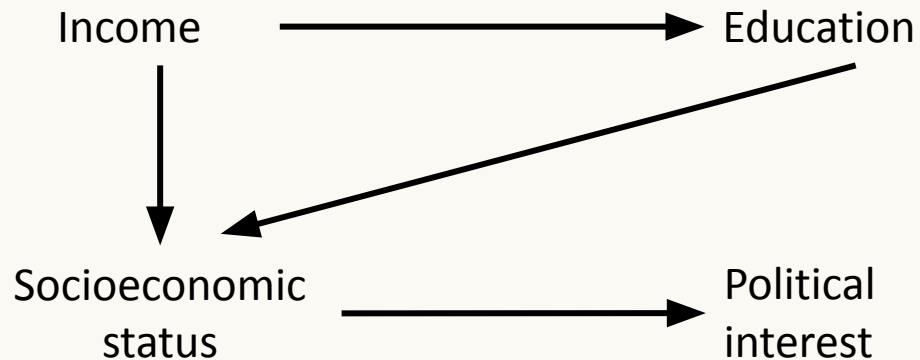
→The second type of research is more complex

Important variables in Descriptive Studies

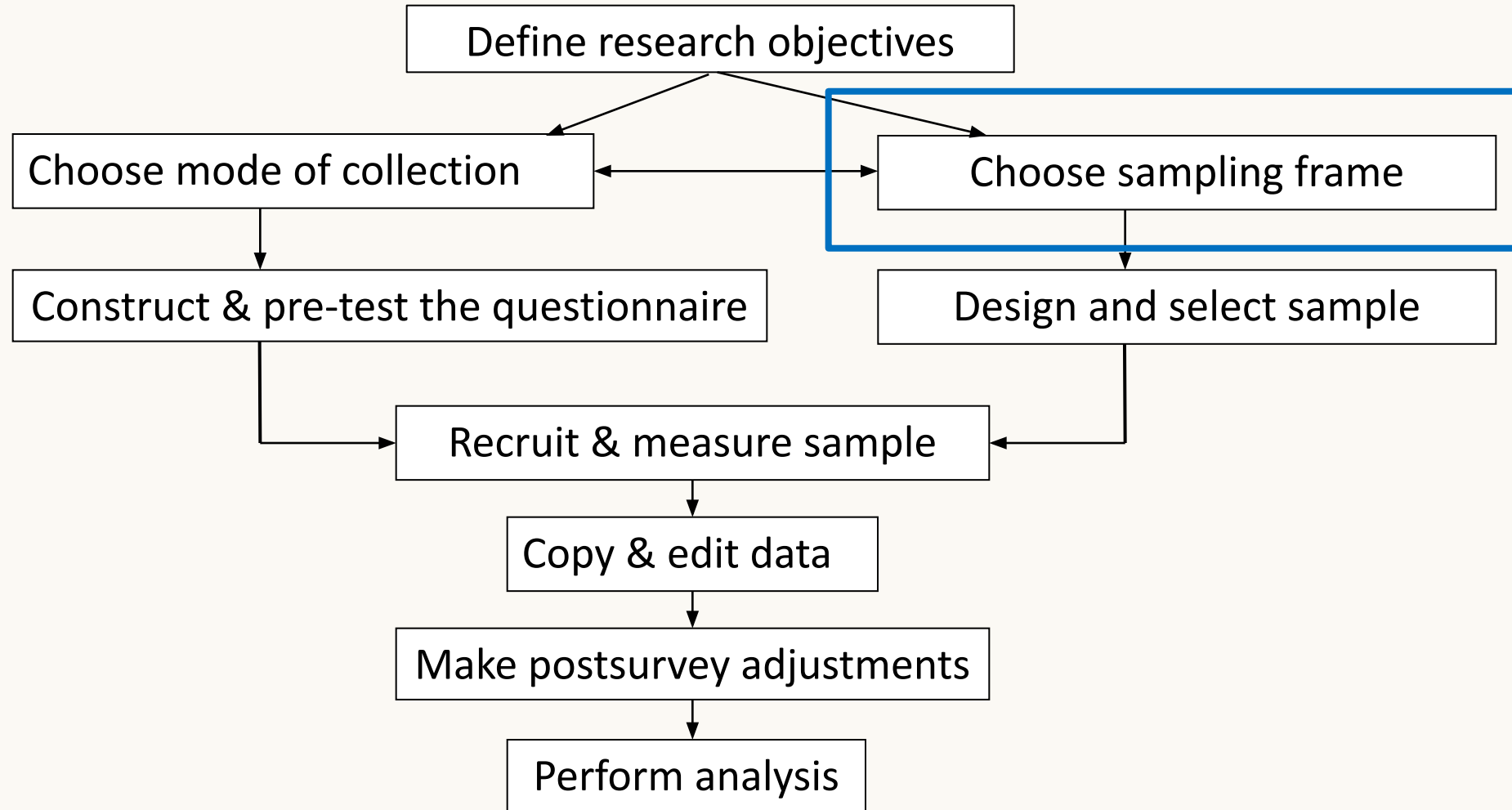
- The choice of variables in descriptive studies seems rather simple
- But often people want to have much more information than only the distribution of the responses for the variable “the opinion about the foreign policy of the US”
- They also want to know who is in favor and who are against. So, they also introduce the variables gender, age , education, income, political orientation, political interest, etc.

Important variables in Explanatory Studies

- Explanatory studies require a more rigorous approach to the selection of variables to determine the effect of different variables.
- Causal models are needed to determine what variables to introduce
- Example: model to determine the effect of Socioeconomic status on political interest



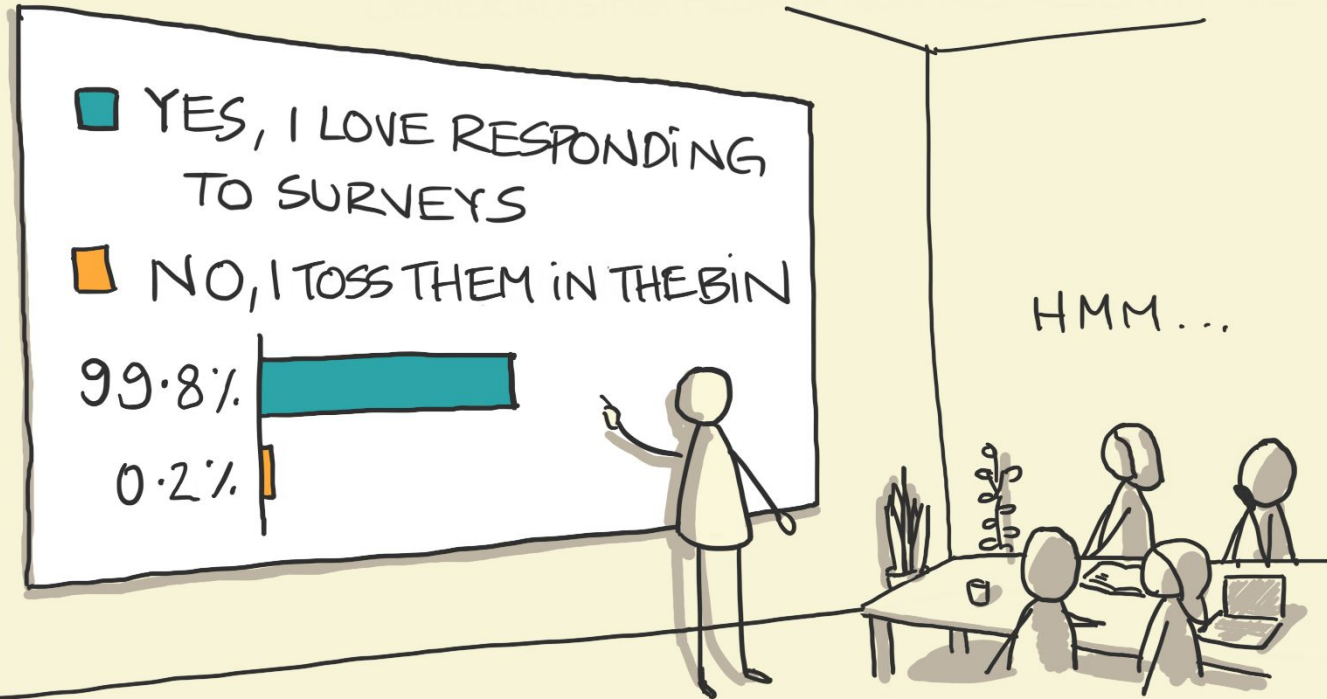
A survey from a process perspective



Who? Target population

- The selection of a population is important because it determines to what population can be generalized after the data collection
→ For questionnaire design: target population determines the use of language, i.e. formal or informal, e.g. Spanish tú/usted, Portuguese tu/você, German du/Sie
- The choice is also not so simple. Do we include:
 - People in institutions like prisons, hospitals, army?
 - People above 80 and below 18?
 - Homeless people?
 - People without legal status/undocumented migrant(s)?
 - Etc. ...

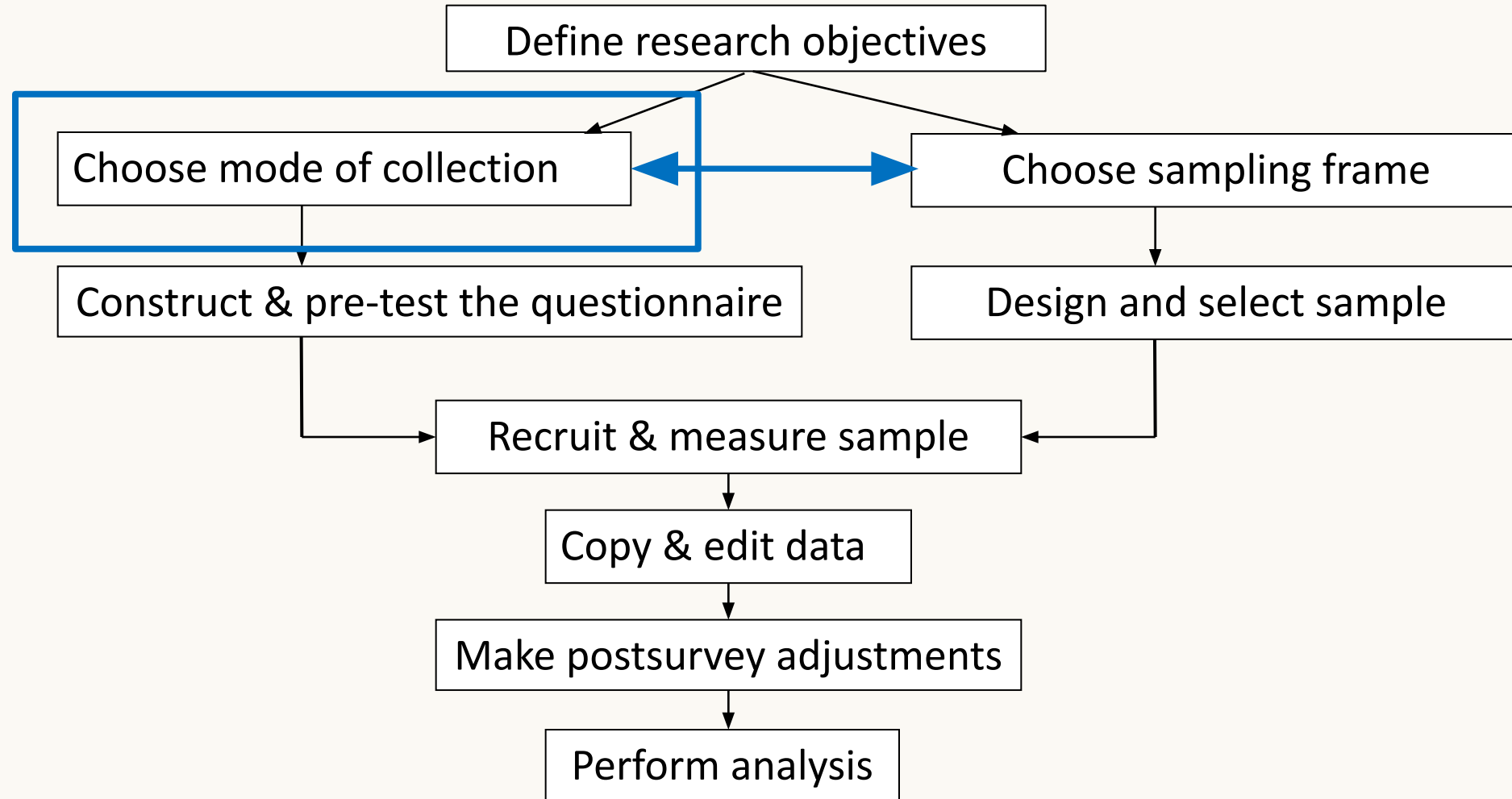
SAMPLING BIAS



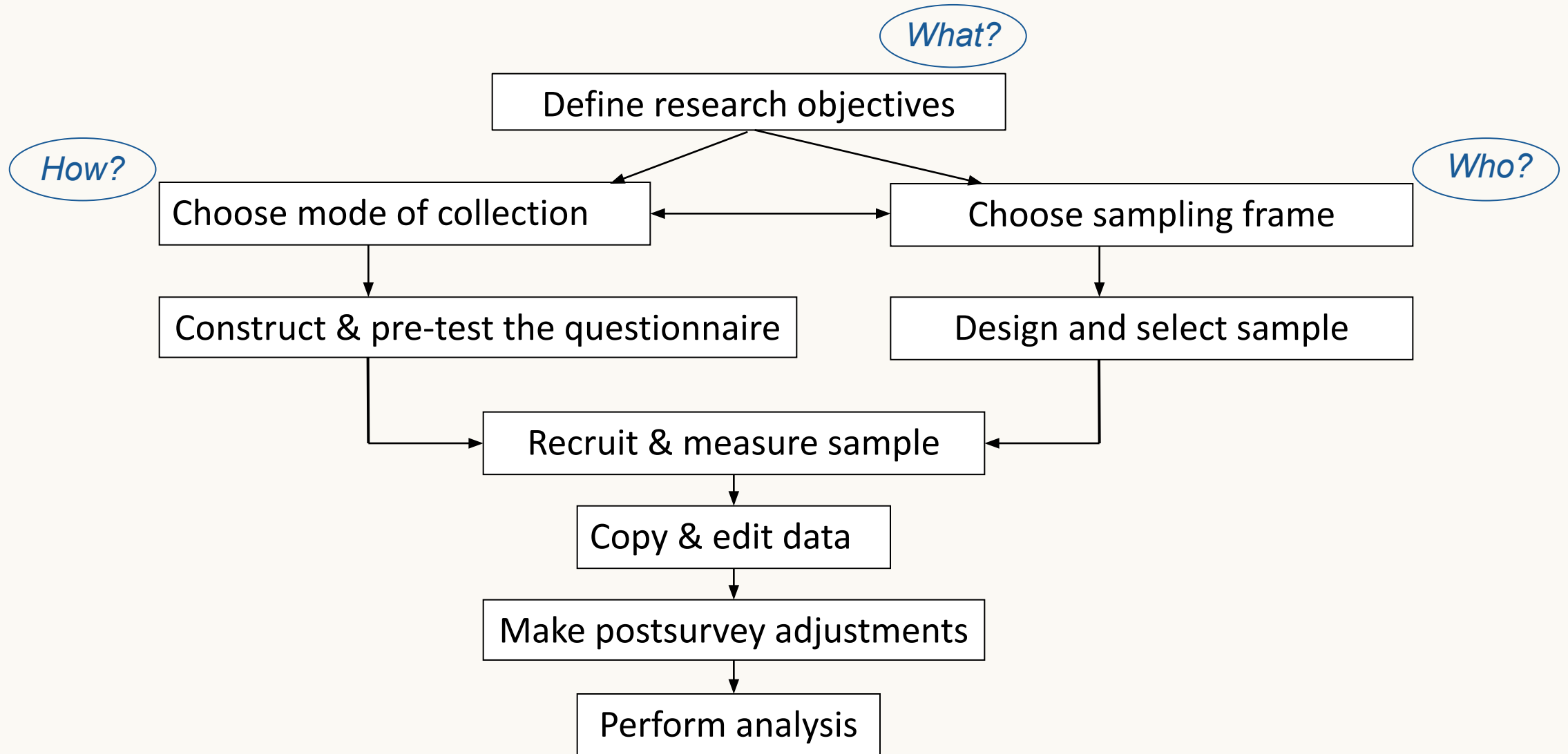
" WE RECEIVED 500 RESPONSES AND
FOUND THAT PEOPLE LOVE RESPONDING
TO SURVEYS "

sketchplanations

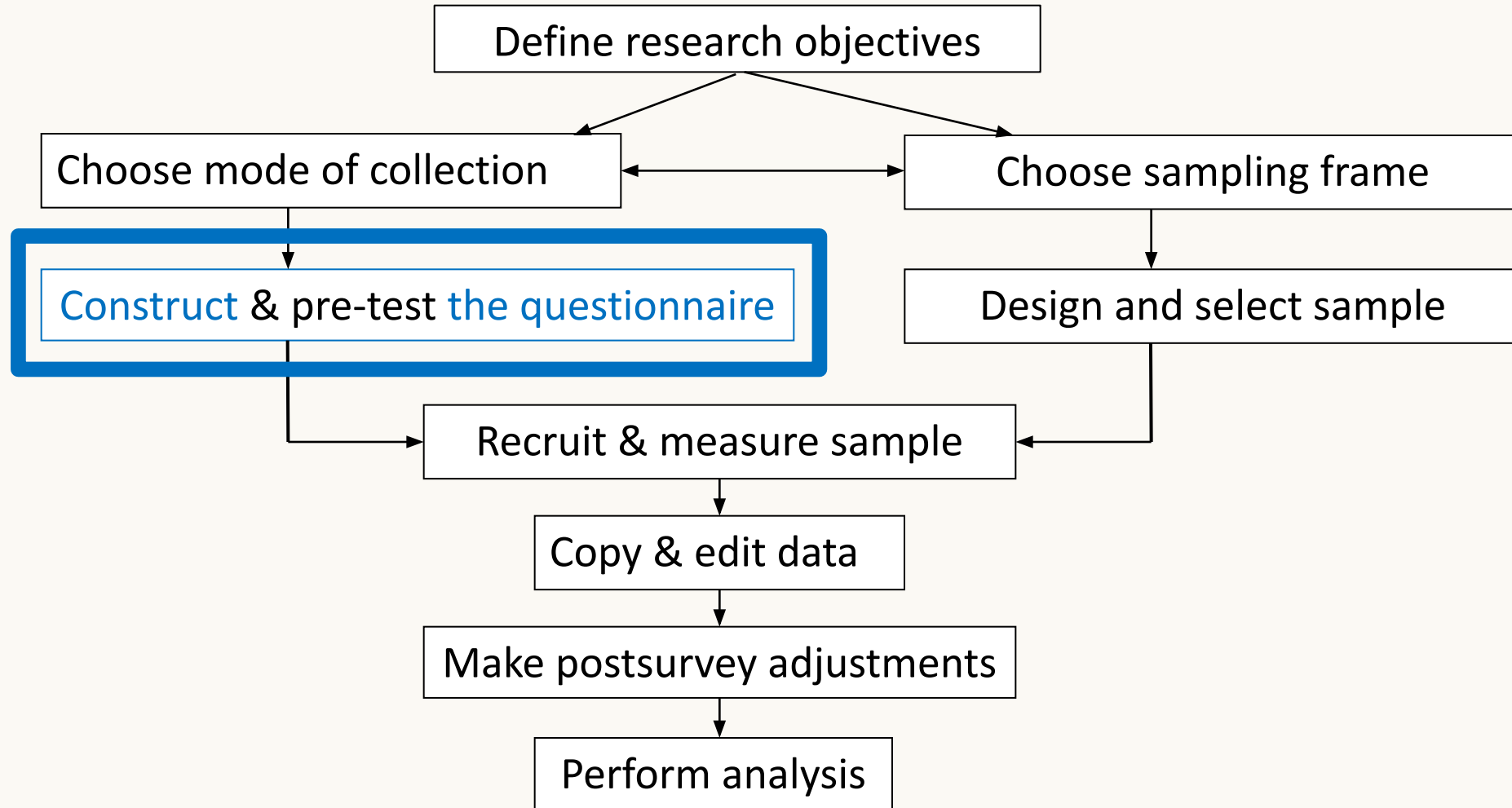
A survey from a process perspective



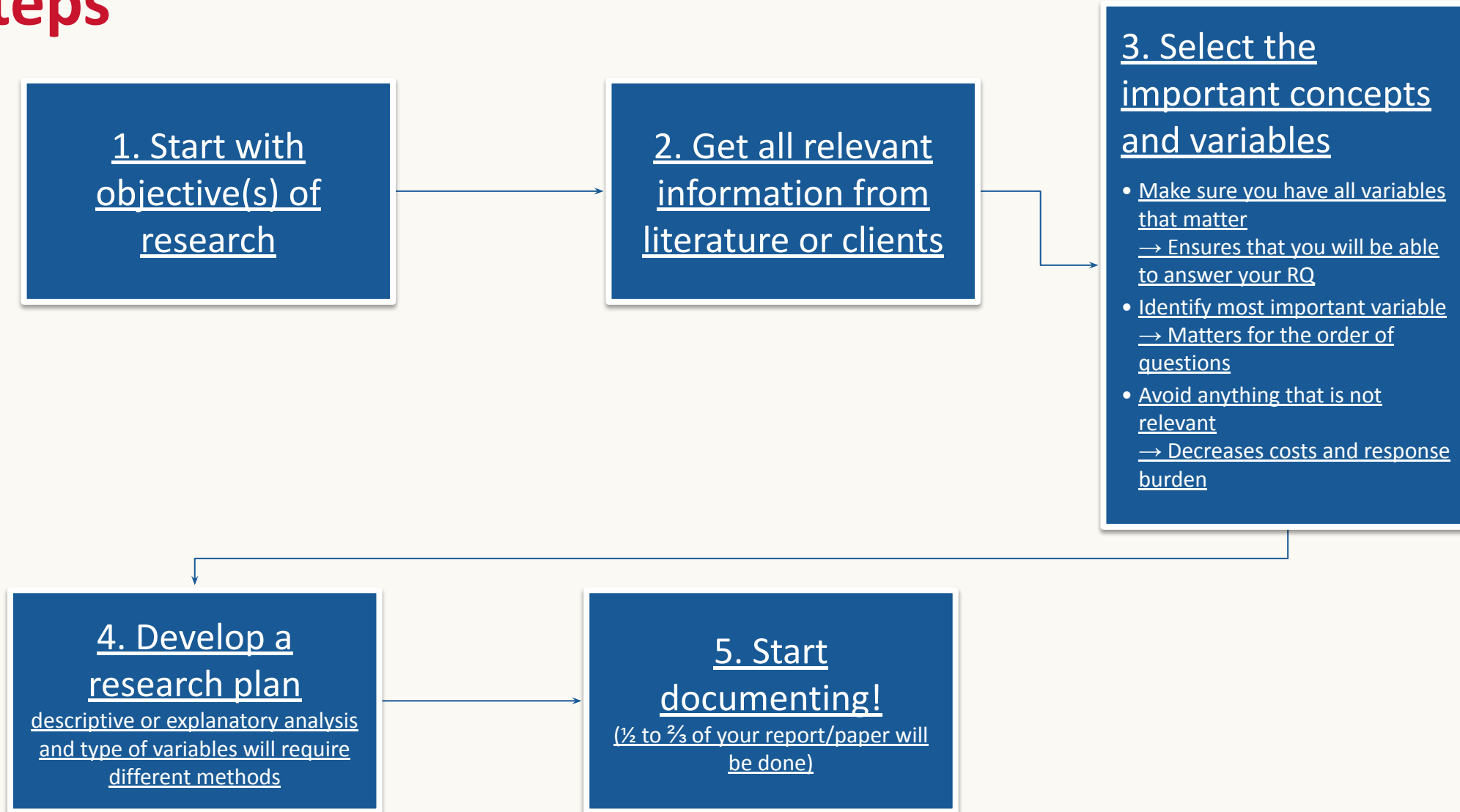
A survey from a process perspective



A survey from a process perspective



Steps



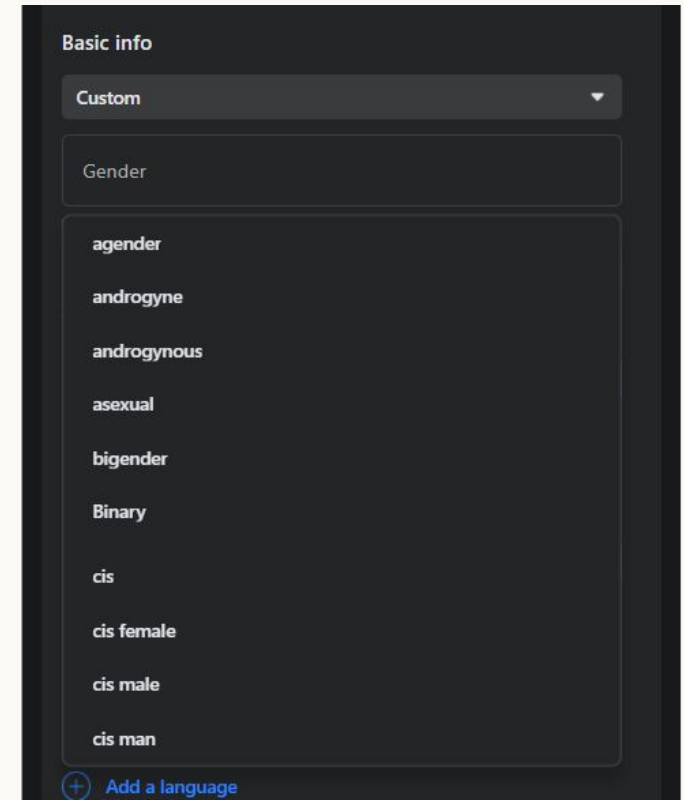
Example: gender



Inclusive intake forms

- **1. What is your current gender identity? (Check and/or circle ALL that apply)**
 - ☐ Male
 - ☐ Female
 - ☐ Transgender Male/Transman/FTM
 - ☐ Transgender Female/Transwoman/MTF
 - ☐ Genderqueer
 - ☐ Additional category (please specify): _____
 - ☐ Decline to answer
- **2. What sex were you assigned at birth? (Check one)**
 - ☐ Male
 - ☐ Female
 - ☐ Decline to answer
- **3. What pronouns do you prefer?** _____ [6]

[\(https://slideplayer.com/slide/14059812/\)](https://slideplayer.com/slide/14059812/)



The image shows a screenshot of a Facebook profile's 'Basic info' section. At the top, there's a dropdown menu set to 'Custom'. Below it is a 'Gender' label. A scrollable list of gender options is shown: agender, androgynous, androgynous, asexual, bigender, Binary, cis, cis female, cis male, and cis man. At the bottom of the list is a blue link that says '+ Add a language'.

Facebook/About/Contact and basic information

Research object:

- 1) Describe fruit consumption
- 2) Explain discrimination

Anna Lindqvist, Marie Gustafsson Sendén & Emma A. Renström (2020): What is gender, anyway: a review of the options for operationalising gender, *Psychology & Sexuality*, DOI: 10.1080/19419899.2020.1729844

Fraser, G. (2018). *Evaluating inclusive gender identity measures for use in quantitative psychological research*. *Psychology & Sexuality*, 1–15. doi:10.1080/19419899.2018.1497693

Smith TW, Son J. Transgender and Alternative Gender Measurement on the 2018 General Social Survey [Internet]. 2019

HOW YOU SHOULD NOT START

Bad practice

What you should not do!

- Start with generating questions from the list of common expressions used
- or
- take questions from former research with the assumption that they worked well*

*The only reason that justifies to use someone else's questions is if you want to compare your study with their study, however, also in this case you should only do that after you have defined your research objectives and key concepts.

Example from practice (1)

Attitudes towards immigrants

- Immigrants should not push themselves where they are not wanted.
- If immigrants would try harder, they could be as well-off as other people.
- How often did you feel sympathy for immigrants?

Example from practice (2)

Question for Policy of Governmental intervention:

Is private enterprise the best way to solve our economic problems?

What is wrong?

Not about “Policy of Governmental intervention”, so it the wrong concept for government intervention

Improvement:

Should the government intervene to solve our economic problems?

→ Not a great request for answer but at least measures “Policy of Governmental intervention”

Consequence

- Evaluating many questionnaires Saris & Gallhofer (2007) have seen that:
 - The “**basic concepts**”* of survey questions are often unclear and sometimes not in agreement with what the questions are supposed to measure
 - They refer to “basic concepts” like evaluation, feeling, importance, norms, or policy (cf. table later)

* in other publications also called “abstract concepts”

How to avoid that?

- First, it is crucial to define very precisely our concepts of interest
 - Example: we want to study if people like football
 - Do we mean that they like playing football?
 - Or that they like watching football?
 - Or both?
- Then, Saris & Gallhofer (2007) have developed a procedure to design questions which gives more certainty about the theoretical validity of the questions: the 3-step procedure

3-STEP PROCEDURE OF SARIS & GALLHOFER

3-step procedure of Saris & Gallhofer

1. Distinction between concepts-by-postulation and concepts-by-intuition
2. Development of assertions for concepts-by- intuition
3. Development of requests for an answer from assertions

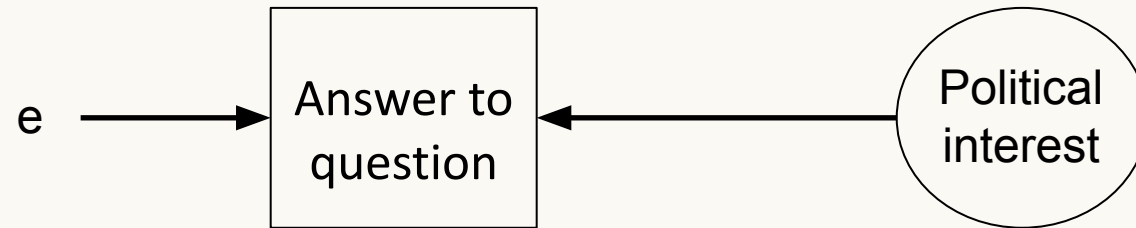
STEP 1:
DISTINCTION BETWEEN
CONCEPTS-BY-POSTULATION (CP) AND
CONCEPTS-BY-INTUITION (CI)

Distinction CP and CI

- Concepts-by-intuition (CI)
 - Simple concepts whose meaning is immediately clear
 - Can be measured with one question

Example: Political Interest measured as CI

How interested are you in politics?



Path diagram:

Oval/circle = latent (unobserved) variable/concept

Square/box = manifest (observed) variable

Distinction Concepts-by-postulation and Concepts-by-intuition

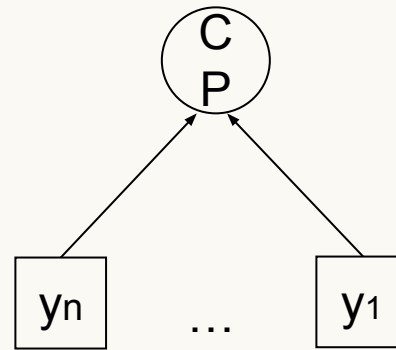
- This distinction following Northrop (1947) and Blalock (1990)
- Concepts-by-intuition (CI)
 - Simple concepts whose meaning is immediately clear
 - Can be measured with one question
- Concepts-by-postulation (CP)
 - Also called “constructs”
 - More complex concepts which require explicit definitions
 - Need more than question to measure them

For concepts-by-postulation (CP)

- Distinction between concepts with
 - **Formative** indicators
 - **Reflective** indicators
- Often, there are many different ways to operationalize the same concept

CP with formative indicators

- Concept defined by a combination of indicators, each indicator measuring a component of the CP
- These variables are not necessarily correlated
- The combination defines the CP



Path diagram:

Oval/circle = latent (unobserved) variable/concept

Square/box = manifest (observed) variable

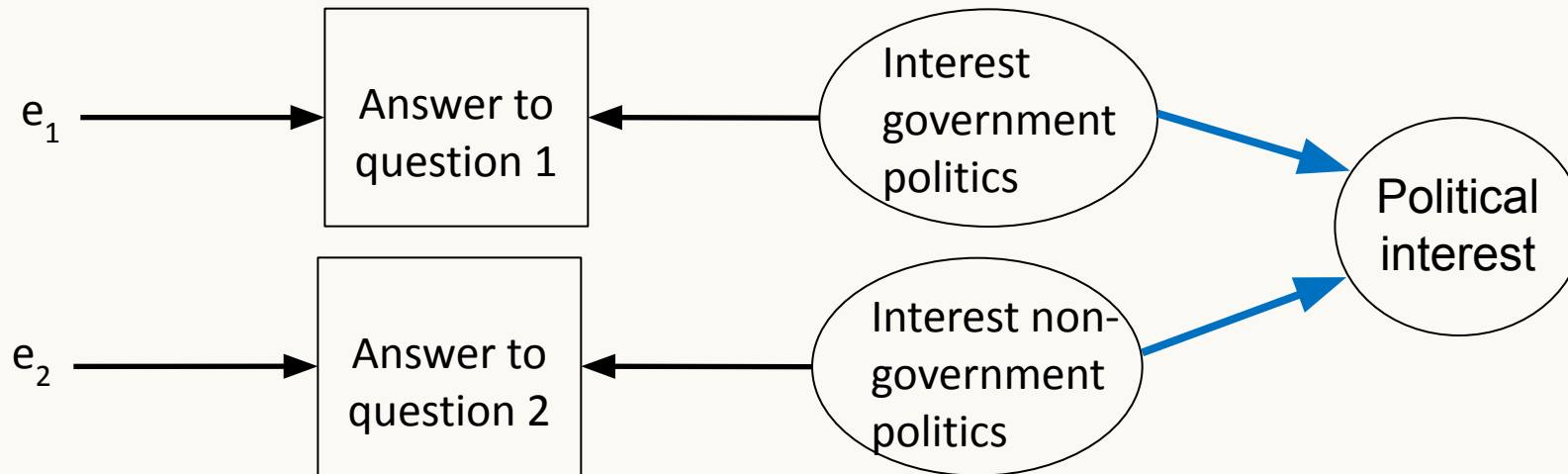
- Requirement:
the definition should take into account all necessary components

Example:

Political Interest measured as CP - formative indicators

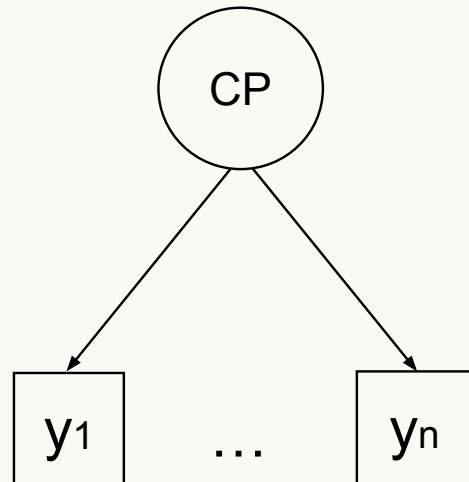
“Political interest” consists of two components:

- Interest in activities of party and government organizations:
 - Question 1: *How interested are you in government politics?*
- Interest in activities of non-government organizations (NGOs)
 - Question 2: *How interested are you in non-government politics?*



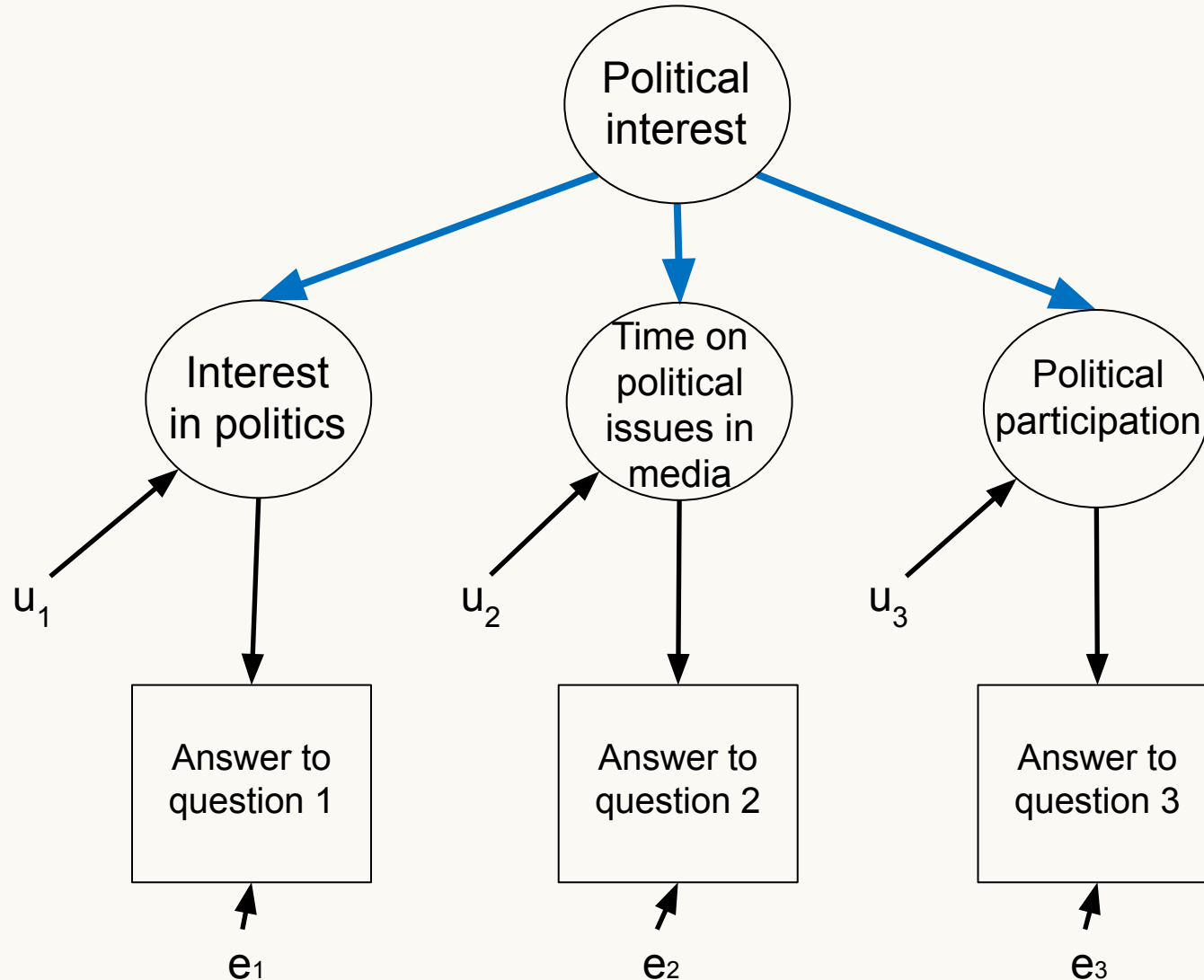
Reflective indicators

- CP is indicated by some similar possible indicators (consequences)
- These indicators will always be correlated because all are affected by the CP
-> If there is a change in CP, all indicators should change consistently.
- Usually 3 indicators are enough (model identified)



Example:

Political Interest measured as CP - reflective indicators

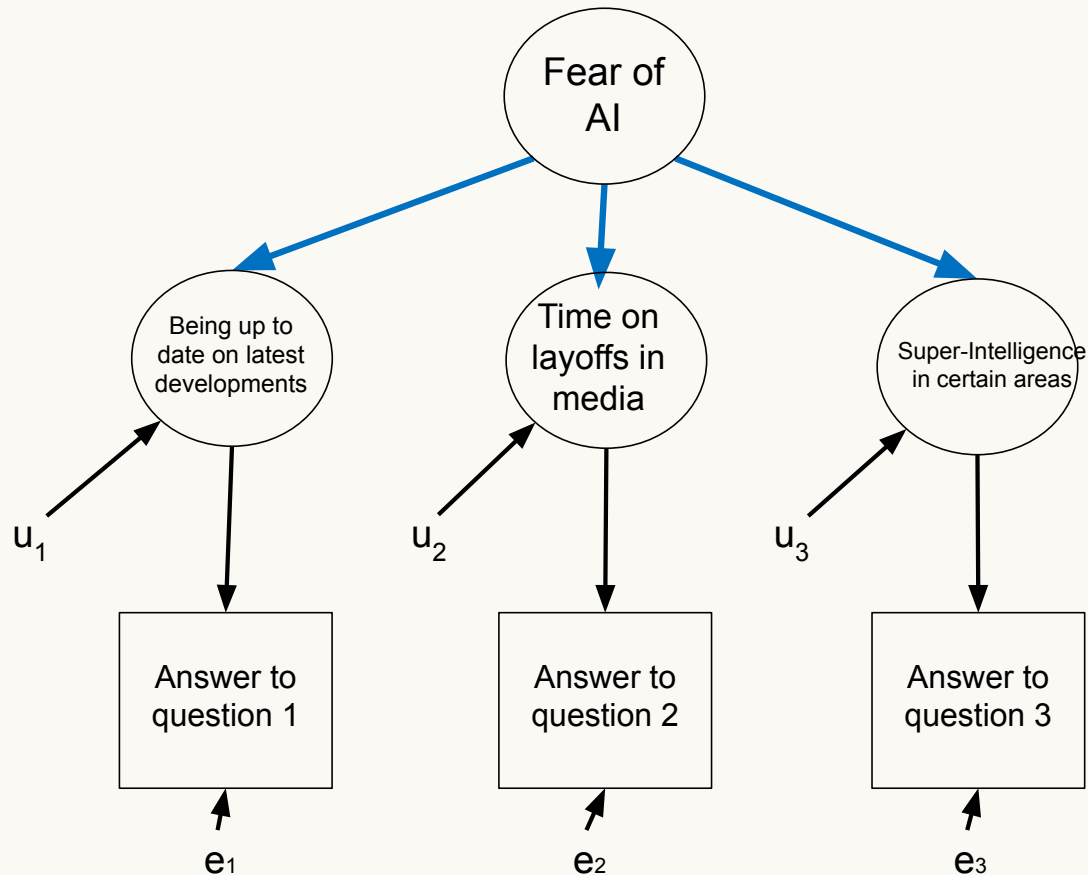


Exercise

1. Do the same with **“Job satisfaction”/“Attitudes towards AI”** or **a concept of your interest**, operationalize it as
 - a CI
 - a CP with formative indicators
 - a CP with reflective indicators
2. Discuss which operationalization is the best - and why?

Group 1: Celal & Leonardo

1. “Fear of losing work to Artificial Intelligence”
2. CP - reflective indicators



Group 2: Vaishali & Somayeh

1. Do the same with “**Job satisfaction**”/“**Attitudes towards AI**” or **a concept of your interest**, operationalize it as
 - a CI
 - a CP with formative indicators
 - a CP with reflective indicators
2. Discuss which operationalization is the best - and why?

Group 3: Lanre & Rudraksha

1. Do the same with “**Job satisfaction**”/“**Attitudes towards AI**” or a concept of your **interest**, operationalize it as

– a CI:

- All things considered, how satisfied are you with your job?

– a CP with formative indicators

- salary
- relationship with coworkers/superior
- distance to office
- working conditions
- growth opportunities at work

– a CP with reflective indicators

- “i am genuinely satisfied with my job”
- “i look forward to going to work”
- “i would recommend this job to other”

2. Discuss which operationalization is the best - and why?

- CP with reflective indicator are the most commonly used (we think) as it enables standard psychometric analysis. It is also robust to small wording changes. The best for JS would be to combine both CP with Formative and reflective together

Group 4: Manfeng & Jiayi

1. Do the same with **“Job satisfaction”/“Attitudes towards AI”** or **a concept of your interest**, operationalize it as
 - a CI: generally, how positive or negative is your attitude towards the use of AI in daily life, work, and society? Using a Likert scale as the response scale, from strongly negative as 1, to strongly positive
 - a CP with formative indicators
 - a CP with reflective indicators
 - [Social benefit] I feel AI are generally beneficial for society?
 - [Comfort with use] I feel comfortable using AI systems in daily life, studies, or work.
 - [Support for use] I support the use of AI systems when

Group 5: Abdul , Vaishali and Alisha

1. Do the same with **“Job satisfaction”/“Attitudes towards AI”** or **a concept of your interest**, operationalize it as
 - a CI
 - Received a promotion in the last two years (0/1)
 - Salary exceeds the industry average (0/1)
 - How satisfy are you with your job (Scale 1-7)
 - a CP with formative indicators
 - How satisfied are you with your salary?
 - How satisfied are you with the management layer at your company?
 - Are there any opportunities for growth?
 - a CP with reflective indicators
 - Are you satisfied with you current position?
 - Do you recommend your workplace to the others?

Summary

- A distinction is made between concepts-by-**postulation** and concepts-by-**intuition**
- All CP can be operationalized using CIs
- For CP we make a distinction between concepts with **reflective** indicators and concepts with **formative** indicators
- Often there are different ways to operationalize concepts

STEP 2:

FROM CONCEPTS TO ASSERTIONS

Formulate assertions for each CI

- We do not go directly to questions because there are too many possibilities
- All CP can be operationalized using CIs, so we only need to formulate the assertions for the CIs

Domains and basic concepts

- There are too many concepts in social sciences
- In order to make rules about them, we first need to **classify** them
- We can classify them according to the list of “**basic concepts**” by Saris and Gallhofer (2007)
- Before formulating the assertions, one should also have clearly in mind the **domain** to which the concepts belong to

Domains and basic concepts

- In the literature, we see concepts like “job satisfaction”, “social trust”, “political participation”, “social contact” etc.
- We can for each of them specify the **domain** (i.e. topic) and classify according to **basic concepts**

	Domain	Basic concept
Job satisfaction	Job	Feeling
Social trust	People	Feeling
Political participation	Politics	Behavior
Attitude towards immigrants	People	Evaluation

Three types of assertions

Linguistic research suggests three assertions' types/structures:

Structure 1:

Subject + link verb predicator + subject complement

X + was + a good President

Structure 2:

Subject + predicator + direct object

I + like + X as a person

Structure 3:

Subject + Predicator

The position of the president + has changed

Assertions and kind of variables

The form of the assertions depends on the kind of variables

- **Subjective variables**

- Information can only be obtained from the respondent (evaluations, judgments, preferences, etc.)

- **Objective variables**

- Information can in principle be obtained by another source: for instance, administrations of towns or hospitals (demographic variables, behavior, etc.)

Assertions for subjective variables

	Structure 1	Structure 2	Structure 3
Basic concepts	$xIsc$	xPy	xP
<i>Subjective variables</i>	<i>I like</i>	<i>like as a person</i>	<i>Position of president has changed</i>
Evaluation	xIe	—	—
Importance	xIi	—	—
Values	vIi	—	—
Feelings	xIf	xFy or xPf	—
Cognitive judgment	xIc	—	—
Causal relationship	$xIca$	xCy	—
Similarity relationship	xIs	xSy	—
Preference	$xIpr$	$xPRy$ (z...)	—
Norms	—	$o(H+I)y$	$o(H+I)$
Policies	—	$g(H+I)y$	—
Rights	$xIri$	$xHRy$	—
Action tendencies	—	$rFDy$	—
Expectations of future events	—	$xFDy$	xFD
Evaluative belief	—	$xP_e y$ or xPy_e	xP_e

Someone/thing (x) is/was(l) good/bad (e)

Someone/thing (x) is/was(l) important (i)

New laws (x) have changed (C) the position of women (y)

Immigrants (o) should (H) adjust to (I) the culture in their new country (y)

Notations: **x** denotes the grammatical subject; **P**, the predicator; **I** the link verb. Frequently occurring subjects are the government (**g**); anyone (**o**); and the respondent him/herself (**r**). Frequently employed lexical verbs for predicators are: **C**, which indicates relationships where the subject causes the object; **D**, which indicates deeds; **E**, which indicates expectations; **F**, which specifies feelings; **FD**, which indicates a predicator referring to future deeds; **H(+I)**, which specifies a predicator which contains words like 'has to' or 'should' followed by an infinitive; **HR**, which specifies predicators like 'has the right to'; **J**, which specifies a judgment; **PR**, which indicates predicators referring to preferences; **S**, which indicates relationships of similarity or difference between the subject and the object.

Revilla, M., D. Zavala-Rojas, and W. Saris (2016). "Creating a good question: How to use cumulative experience". In C. Wolf, D. Joye, T.W. Smith and Y.-C. Fu (editors), *The SAGE-Handbook of Survey Methodology*. SAGE. ISBN: 9781446282663. Chapter 17, page 240

From the table to assertions

- Using the **same structure** and **domain** we can formulate assertions for **different basic concepts**:
- Subject: My work

Evaluation: My work is **useful**
Importance: My work is **important**
Feeling: My work is **pleasant**
Cognition: My work is **physical**

Basic concepts	Structure 1 xIsc
<i>Subjective variables</i>	
Evaluation	xIe
Importance	xIi
Feelings	xIf
Cognitive judgment	xIc

Different assertions for the same concept

Basic concept: feeling. Domain: politics

	Structure 1	Structure 2	Structure 3
Basic concepts	xIsc	xPy	xP
Feelings	xIf	xFy or xPf	—

- Feeling structure 1:
Subject + link verb predicator + Subject complement
Politics + are + annoying
- Feeling structure 2:
Subject + predicator + direct object
Politics + annoys + me

From the assertions to the table?

- What basic concept is represented in these assertions?
 - “Many other groups have come here and overcame prejudice and worked their way up. Immigrants should do the same without demanding special favors.”
 - “Over the past few years immigrants have received less than they deserve.”

Use of these production rules

- Using these production rules leads to a formulation of an assertion which represents the concept of interest.
- Using the table in the opposite direction (from a question to a concept) is not so simple because researchers sometimes make very confusing sentences

Objective variables

- The same procedure can be used for objective variables
- In this case, there are usually less confusion, but it does not harm to use the same procedure

Assertions for objective variables

	Structure 1	Structure 2	Structure 3
Basic concepts	xIsc	xPy	xP
<i>Objective variables</i>			
Behavior	—	rDy	rD
Events	—	xDy	xD
Demographics	xId	—	—
Knowledge	xIsc	xPy	xP
Time	—	—	xDti
Place	—	—	xDpl
Quantities	—	xDqu	—
Procedures	—	—	xDpl, pro

Examples of Objective variables

- Structure 1: Demographic variable

Subject + link verb predicator + subject complement

I + am + a scientist

- Structure 2: Behavior

Subject + predicator + direct object

The president + has denied + the use of force

- Structure 3: Behavior

Subject + predicator

I + worked yesterday

Summary

- For each CI, we should specify the domain and classify it according to the basic concept
- Then, we use the rules in the tables to formulate assertions which measure with certainty the concept of interest:
Saris, W. and I. Gallhofer. 2014. Design, Evaluation, and Analysis of Questionnaires for Survey Research. Hoboken, NJ: John Wiley & Sons.
Page 50.

STEP 3:
FROM ASSERTIONS TO
REQUESTS FOR ANSWERS

From assertions to requests for an answer

Assertion: X is a good prime minister

Request for an answer:

1. Interrogative (question)

Is X a good prime minister?

2. Imperative (orders, instructions)

Please, tell me whether X is a good prime minister.

3. Declarative (assertion) – no request but answer expected

I would like to ask you whether X is a good prime minister.

Different structures for requests for answers

- WH words: who, whom, whose, what, which, when, where, why, how
- Direct or indirect request

WH words	Direct request	Indirect request	
WH word not present	Direct instruction	Imperative + interrogative	—
	Direct request	Interrogative + interrogative	Interrogative + declarative
	—	Declarative + interrogative	—
WH word present	—	Imperative + interrogative	—
	Direct request	Interrogative + interrogative	—
	—	Declarative + interrogative	—

Different structures for requests for answers

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	—	Declarative + interrogative	—
WH word present	—	Imperative + interrogative	—
	Direct request	Interrogative + interrogative	—
	—	Declarative + interrogative	—

Direct requests without WH words

- **Direct instruction/imperative**

Please, indicate your opinion about the functioning of the economy using a number between 0 and 100, where 0 indicates very bad and 100 very good.

- **Direct request/interrogative**

Are you going to vote during the next elections?

Different structures for requests for answers

- WH words: who, whom, whose, what, which, when, where, why, how
- Direct or indirect request

WH words	Direct request	Indirect request	
WH word not present	Direct instruction	Imperative + interrogative	—
	Direct request	Interrogative + interrogative	Interrogative + declarative
	—	Declarative + interrogative	—
WH word present	—	Imperative + interrogative	—
	Direct request	Interrogative + interrogative	—
	—	Declarative + interrogative	—

Direct interrogative with a WH word

Why is X a good prime minister?

How good is X as a prime minister?

→ *typically avoid leading or unbalanced request*

Different structures for requests for answers

- WH words: who, whom, whose, what, which, when, where, why, how
- Direct or indirect request

WH words	Direct request	Indirect request	
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	—	Declarative + interrogative	—
WH word present	—	Imperative + interrogative	—
	Direct request	Interrogative + interrogative	—
	—	Declarative + interrogative	—

Indirect requests

Indirect requests consist of two parts:

1. Pre request (main clause)

Indicates that information about something is required

2. Subordinate clause

Indicates what one wants to know

Indirect requests

Assertion

X is a good prime minister

- **Indirect imperative**

Pre request + subordinate clause

Can you tell me + to what extent X is a good prime minister.

- **Indirect interrogative**

Pre request + subordinate clause

Do you think + that X is a good prime minister?

Exercise

- Assertion:

“I am satisfied with my job”, “I am positive about AI.” or your a assertion of your concept of interest.

Transform it in a request of the following forms:

- Direct interrogative with or without a WH word
- Direct imperative
- Indirect interrogative
- Indirect imperative

Exercise Group 1: Manfeng & Jiayi

- Assertion:

“I am satisfied with my job”, “I am positive about AI.” or your a assertion of your concept of interest.

Transform it in a request of the following forms:

- Direct interrogative with or without a WH word
- Direct imperative
- Indirect interrogative
- Indirect imperative

Exercise Group 2: Celal & Leonardo

- Assertion:

“I am satisfied with my job”, “I am positive about AI.” or your a assertion of your concept of interest.

Transform it in a request of the following forms:

- Direct interrogative with or without a WH word: *Are you satisfied with your job?*
- Direct imperative: *Tell me if you are satisfied with your job.*
- Indirect interrogative: *Can you let me know whether you are satisfied with your job?*
- Indirect imperative: *I'd like you to consider your job satisfaction.*

Exercise Group 3: Vaishali & Somayeh

- Assertion:

"I am satisfied with my job"

Transform it in a request of the following forms:

- Direct interrogative with or without a WH word:

- *"Are you satisfied with your job?"*

- Direct imperative:

1. *"Rate your level of satisfaction with your job."* or
2. *"Indicate how satisfied you are with your job."*

- Indirect interrogative:

- *"We would like to know how satisfied you are with your job."*

- Indirect imperative:

- *"Respondents are requested to consider how satisfied they feel with their current job."*

Exercise Group 4: Abdul and Alisha

- Assertion:

“I am satisfied with my job”, “I am positive about AI.” or your a assertion of your concept of interest.

Transform it in a request of the following forms:

- Direct interrogative with or without a WH word: Are you positive about AI?
- Direct imperative: Please indicate how positive you are about AI.
- Indirect interrogative: Could you tell us how positive you are about AI?
- Indirect imperative: Could you describe your opinion about AI?

Exercise Group 5: Lanre & Rudraksha

- Assertion:

“I am satisfied with my job”, “I am positive about AI.” or your a assertion of your concept of interest.

Transform it in a request of the following forms:

- Direct interrogative with or without a WH word
 - are you positive about AI?
 - how positive are you about AI? (with WH)
- Direct imperative
 - rate how positive you feel about AI
- Indirect interrogative
 - i would like to know if you feel positive about AI?
- Indirect imperative
 - could you tell me how positive you feel about AI?

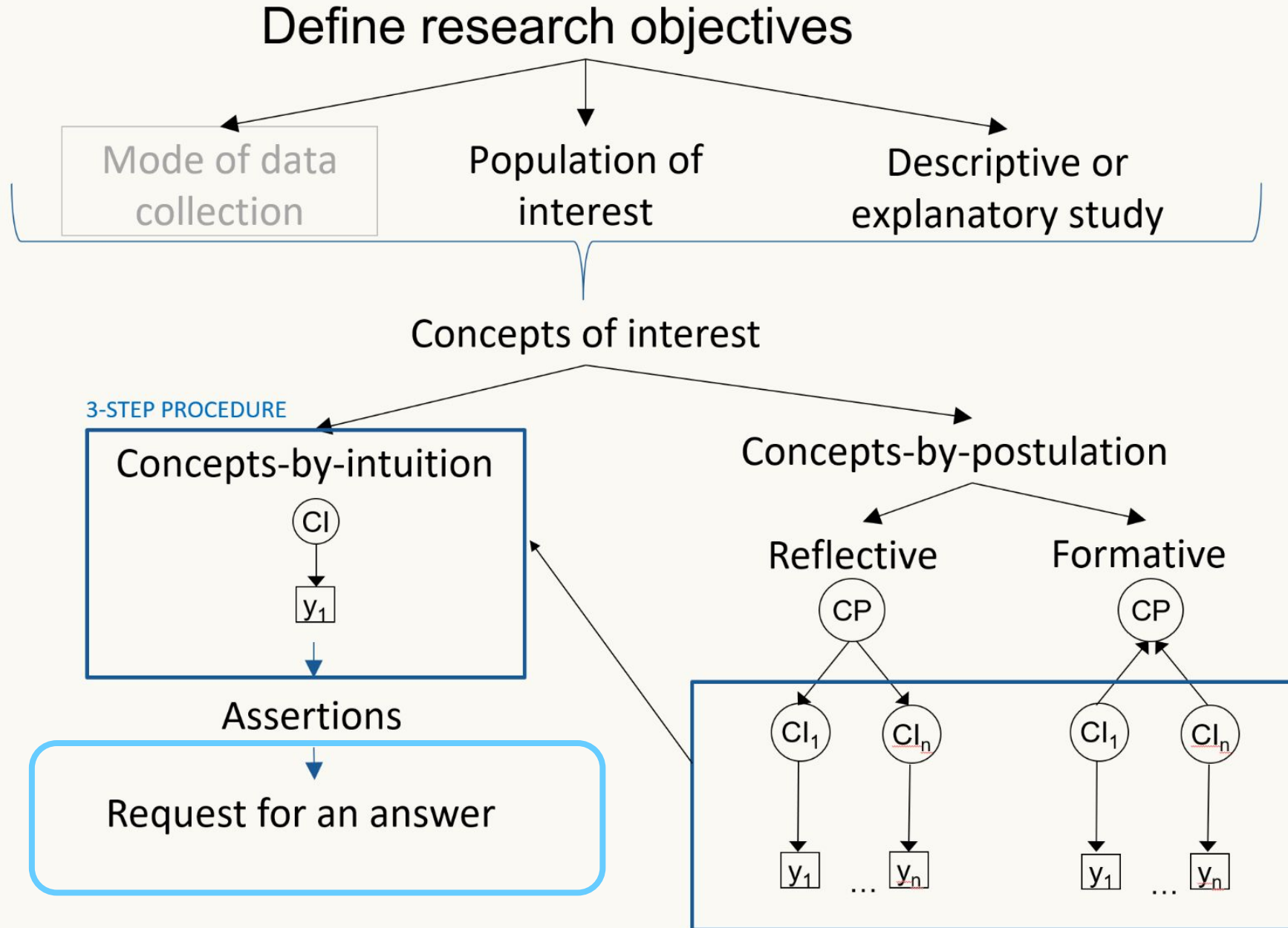
Questionnaire design

- Part 2 -

Wiebke Weber
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LMU Munich
18 May 2026

Summary from last session



RESPONSE PROCESS

Response process: four steps

1. Comprehension



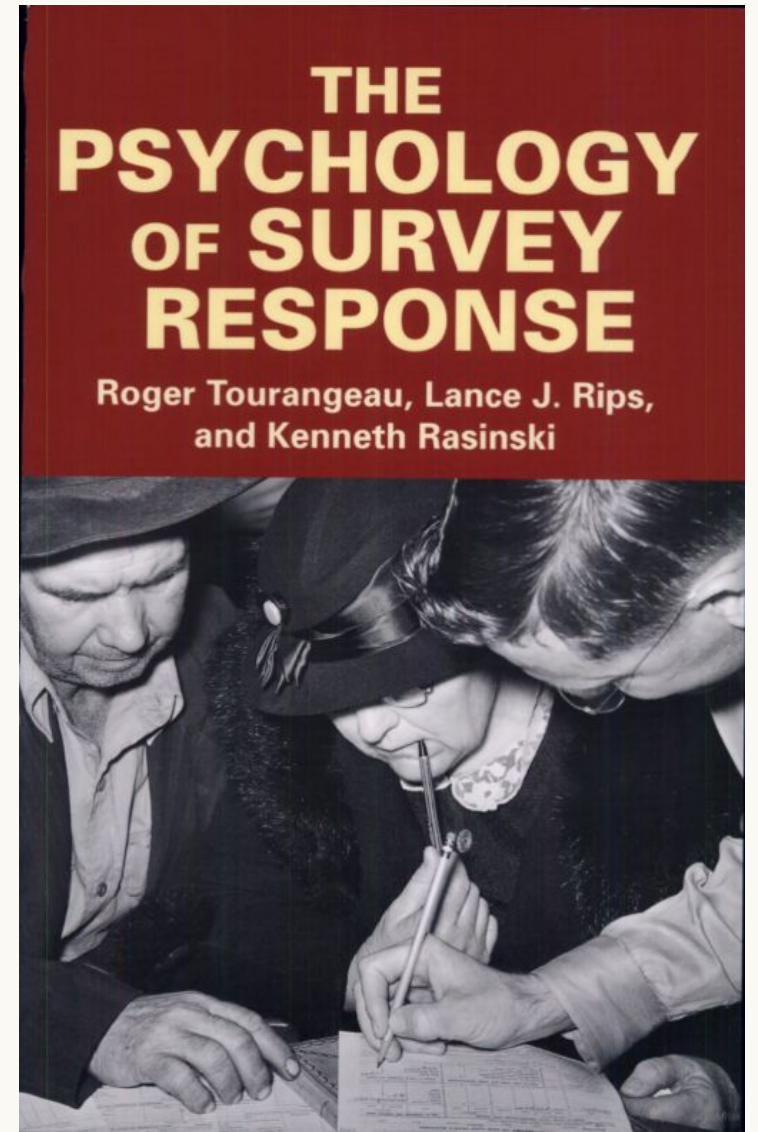
2. Retrieval



3. Judgement



4. Response



1.Comprehension



- understand/interpret the intended meaning of question
 - Knowledge of words/sentences – literal understanding
 - but also context of the question in the questionnaire, the questionnaire in a study, the study in a culture/country, ...

2. Retrieval



- search memory for all relevant information
- Generate retrieval strategy and cues
- Retrieve specific, generic memories
- Fill in missing details

3. Judgement



- involves integrating retrieved info into summary judgement
- Assess completeness and relevance of memories
- Draw inferences based on accessibility of memories/information
- Decide on willingness to disclose material retrieved
- Make estimate based on partial retrieval

4. Response



→ translate the judgement into the required format

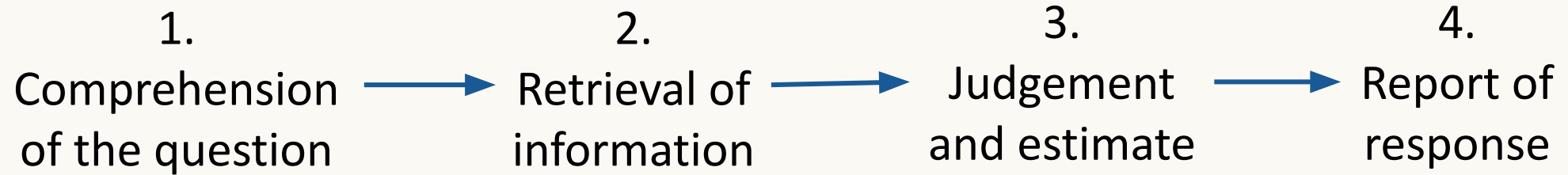
- Map judgment on response category
- Edit response

The screenshot shows a survey titled "Please evaluate your stay" with four categories: Housekeeping, Reception services, Restaurant, and Decoration. Each category has a horizontal Likert scale bar with five segments. The first two segments are blue, and the third segment contains a yellow sad face emoji. A mouse cursor is hovering over the end of the Housekeeping scale. At the bottom, there are "PREVIOUS" and "NEXT" navigation buttons.

Category	Scale (1 to 5)
Housekeeping	Blue Blue Yellow Sad Face Grey Grey
Reception services	Blue Blue Yellow Sad Face Grey Grey
Restaurant	Blue Blue Yellow Sad Face Grey Grey
Decoration	Blue Blue Yellow Sad Face Grey Grey

<https://s3.amazonaws.com/www-inside-design/uploads/2018/06/likert-2.gif>

Optimizing and satisficing



Proceeding carefully through the 4 step is called “**optimizing**”

→ Is that what most respondents do?

No, most respondents use shortcuts, they “**satisfice**”:

Weak satisficing: superficially engage in 2.retrieval +3.judgement

Strong satisficing: skip 2.+3. completely, respondents cue from question and situation for apparently plausible answers

Reasons for satisficing

- Respondent abilities & task difficulty
 - ☐ Target population: expert vs. general survey
 - ☐ Only ask the essential for your research objectives
 - ☐ Make questions easy
- Respondent motivation
 - ☐ Explain importance in introduction
 - ☐ Use incentives
- Time
 - Time pressure (perceived or real as you have to go)
 - ☐ Survey length: only ask the essential, as short as possible

Satisficing behaviour

- Selecting the first reasonable response:
 - Visual presentation: primacy effect (first seen)
 - Oral presentation: recency effect (most recent heard)
- Agreeing with assertions independently from their content = acquiescence (yes-saying/agreeing)
- Disagreeing with assertions independently from their content (no-saying)

Satisficing behaviour

- Non-differentiation (straightlining) in ratings:

Extreme answers

Avoiding extreme answers

Midpoint answers

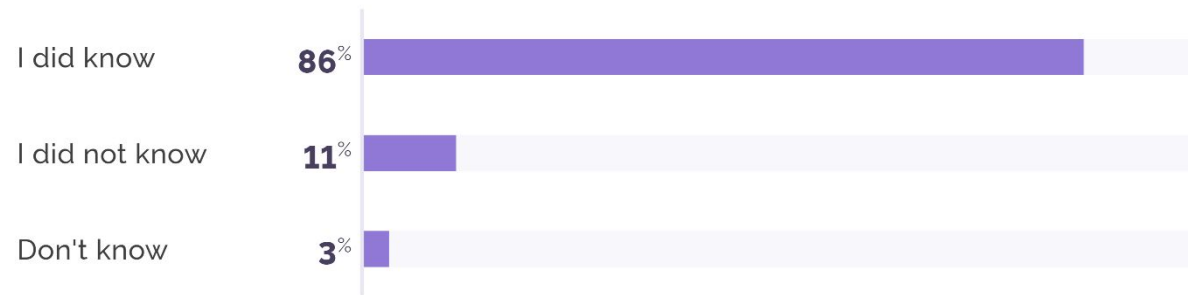
Very dissatisfied 1	Dissatisfied 2	Neither satisfied nor dissatisfied 3	Satisfied 4	Very satisfied 5
F18.3 (F23.2.3)How satisfied are you with your financial situation?				
Very dissatisfied 1	Dissatisfied 2	Neither satisfied nor dissatisfied 3	Satisfied 4	Very satisfied 5
F19.3 (F24.2.1)How satisfied are you with your access to health services?				
Very dissatisfied 1	Dissatisfied 2	Neither satisfied nor dissatisfied 3	Satisfied 4	Very satisfied 5
F19.4 (F24.2.5)How satisfied are you with the social care services?				
Very dissatisfied 1	Dissatisfied 2	Neither satisfied nor dissatisfied 3	Satisfied 4	Very satisfied 5
F20.3 (F25.2.1)How satisfied are you with your opportunities for acquiring new skills?				
Very dissatisfied 1	Dissatisfied 2	Neither satisfied nor dissatisfied 3	Satisfied 4	Very satisfied 5
F20.4 (F25.2.2)How satisfied are you with your opportunities to learn new information?				
Very dissatisfied 1	Dissatisfied 2	Neither satisfied nor dissatisfied 3	Satisfied 4	Very satisfied 5
F21.4 (F26.2.3)How satisfied are you with the way you spend your spare time?				
Very dissatisfied 1	Dissatisfied 2	Neither satisfied nor dissatisfied 3	Satisfied 4	Very satisfied 5

Satisficing behaviour

- Don't know answers
- Mental coin-flipping

Did you know that a new Labour party leader was elected this week?

All adults (2763 GB adults - Apr 8, 2020)



YouGov® | What the world thinks

yougov.com

<https://yougov.co.uk/topics/politics/survey-results/daily/2020/04/08/ad558/2>

REQUEST FOR AN ANSWER

Decomposition of a survey item

1. Introduction of the topic
2. Motivation for the respondent to answer
3. Information regarding the content
4. Information regarding a definition
5. Instruction for the respondent
6. Instruction for the interviewer
- 7. Request for an answer** CORE UNIT; in its simplest form
it's an open request
8. Answer categories or response scales

Request for answer - Connected with research objectives

1. Time reference
2. Saliency or centrality
3. Social desirability
4. Sensitive topics

1. Time reference: present & future

- Present: feeling at the moment
satisfaction with different aspects of life
opinion about politics
- Future: buying a product
will participate in an event
expect change or events
→ **not a far-off future**

1. Time reference: past

- Past:

the only reference period, that gives some choice

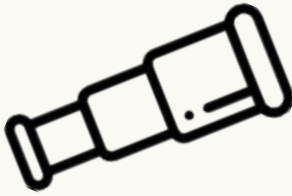
→ the longer the reference period, the more unlikely that respondent remember, in particular for activities that occur frequently

Example:

How much time did you spend watching programs on politics or actuality last week?

How much time did you spend watching programs on politics or actuality yesterday?

1. Time reference: past - telescoping



- Telescoping: memory/recall error in temporal placement of events
 - Forward: reporting of events as being more recent than they actually were
 - Backward: reporting of events as farther back than they actually occurred

- Example (Scherpenzell (1995) in Saris&Gallhofer (2014: 82):

Have you experienced robbery or theft during the last year?

Reported number twice as high than asking:

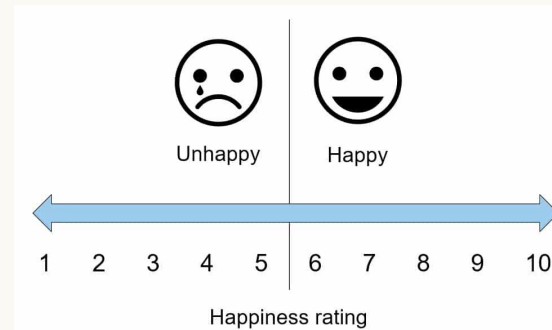
Have you experienced robbery or theft during the last 5 years? How about the last year?

2. Saliency or centrality

- saliency of opinions depends on the topic asked and the interest people have in this topic

Examples:

How happy are you today?



<https://www.trackinghappiness.com/happiness-scale-from-1-to-10/>

The European Commission is planning to change the “REGULATION (EC) No 216/2008 regarding the European Aviation Safety Agency. To what extent do you support this plan?

→ Give additional information



3. Social desirability

= Respondents answering in a way that they think is viewed favourable by others or in agreement with the social norm

- Lead to under- and over-reporting:

Undereporting:

Criminal behaviour
Alcohol & drug use
Mental health issues
TV viewing
Bad diet
Sexual partners by ♀
Abortion

Overreporting:

Reading books
Visiting museums/galleries
Voting
Volunteering & donating to charity
Exercise
Good diet
Sexual partners by ♂

→ usually less prominent in self-completed surveys

Reducing social desirability : Example income

Example [ESS R8 UK](#)

F41) Using this card, please tell me which letter describes your household's total income, after tax and compulsory deductions, from all sources? If you don't know the exact figure, please give an estimate. Use the part of the card that you know best: weekly, monthly or annual income.

CARD 44				
YOUR <u>HOUSEHOLD</u> INCOME				
	Approximate WEEKLY	Approximate MONTHLY	Approximate ANNUAL	
J	Less than £229	Less than £994	Less than £11,926	J
R	£229 to under £308	£994 to under £1,340	£11,926 to under £16,083	R
C	£308 to under £385	£1,340 to under £1,673	£16,083 to under £20,077	C
M	£385 to under £469	£1,673 to under £2,038	£20,077 to under £24,453	M
F	£469 to under £562	£2,038 to under £2,444	£24,453 to under £29,322	F
S	£562 to under £667	£2,444 to under £2,899	£29,322 to under £34,793	S
K	£667 to under £795	£2,899 to under £3,456	£34,793 to under £41,467	K
P	£795 to under £982	£3,456 to under £4,267	£41,467 to under £51,204	P
D	£982 to under £1,267	£4,267 to under £5,507	£51,204 to under £66,084	D
H	£1,267 or more	£5,507 or more	£66,084 or more	H

→ Offer alternative
response options:
here generic (J, R, C,...)
answer categories

Reducing social desirability: Example alcohol consumption

Example ESS R7 (2014) UK

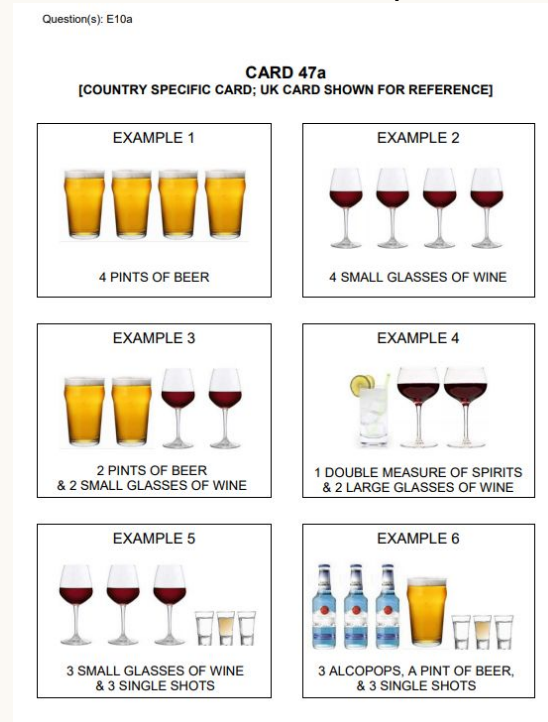
E10a CARD 47a This card shows six different examples of how much alcohol a person might drink on a single occasion.

INTERVIEWER PAUSE TO ALLOW RESPONDENT TO CONSIDER THE SHOWCARD.

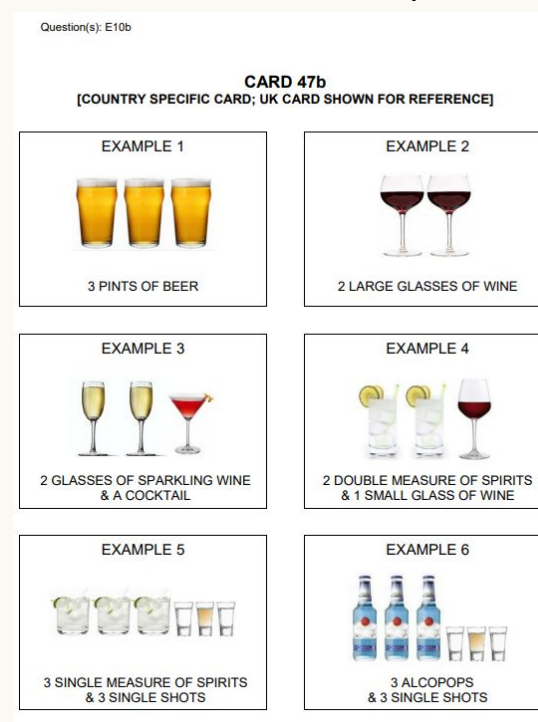
In the last 12 months, how often have you drunk this amount of alcohol or more on a single occasion? Was it... **READ OUT...**

...daily or almost daily,
weekly,
monthly,
less than monthly,
or, never?

Showcard for male respondents



Showcard for female respondents



→ Offer alternative
request for answer:
here images

ESS R7 measure of alcohol consumption

- new method for measuring alcohol consumption
- followed the WHO approach by measuring three concepts:
 - (i) the frequency of alcohol consumption,
 - (ii) the quantity of alcohol consumed, and
 - (iii) binge drinking.
- for (iii), presented typical drinks on a country-specific show card (rather than asking units consumed or number of drinks consumed)
- calculated the likely grams of alcohol in those to enable an overall total to be computed for each respondent

ESS R7 binge drinking – showcard Spain

For male respondents:

Pregunta: E10a

TARJETA 47a

Ejemplo 1

5 tercios



Ejemplo 2

5 copas de vino



Ejemplo 3

1 caña, 2 tercios y 2 copas de vino



Ejemplo 4

3 cañas y 2 combinados



Ejemplo 5

2 cañas, 2 chupitos, 1 cóctel y 1 combinado



Ejemplo 6

1 tercio, 1 copa de vino, 1 combinado y 1 copa de licor fuerte



For female respondents:

Pregunta: E10b

TARJETA 47b

Ejemplo 1

1 caña y 3 tercios



Ejemplo 2

4 copas de vino



Ejemplo 3

1 caña, 1 tercio y 2 copas de vino



Ejemplo 4

1 caña y 2 combinados



Ejemplo 5

1 caña, 1 chupito y 2 cócteles



Ejemplo 6

2 cañas, 1 copa de vino y 1 combinado



ESS R7 binge drinking – showcard Norway

E10a-b This card shows six different examples of how much alcohol a person might drink on a single occasion. In the last 12 months, how often have you drunk this amount of alcohol or more on a single occasion? Was it...(READ OUT)

Daily or almost daily, Weekly, Monthly, Less than monthly, Never?

For male respondents:

KORT 47a

EKSEMPEL 1



5 SMÅ FLASKER ELLER BOKSER
ØL

EKSEMPEL 2



5 SMÅ GLASS VIN

EKSEMPEL 3



2 HALVLITERE ØL OG 2 SMÅ
GLASS VIN

EKSEMPEL 4



2 ENKLE DRINKER
OG
3 SMÅ GLASS VIN

EKSEMPEL 5



1 STORT GLASS VIN
OG 4 ENKLE DRINKER

EKSEMPEL 6



2 RUSBRUS, 1
HALVLITER ØL OG 2
ENKLE DRINKER

For female respondents:

KORT 47b

EKSEMPEL 1



4 SMÅ FLASKER ELLER BOKSER
ØL

EKSEMPEL 2



4 SMÅ GLASS VIN

EKSEMPEL 3



2 SMÅ GLASS VIN OG 2 ENKLE
DRINKER

EKSEMPEL 4



4 ENKLE DRINKER

EKSEMPEL 5



2 HALVLITERE ØL OG 1 ENKEL
DRINK

EKSEMPEL 6



4 RUSBRUS OG
1 ENKEL DRINK

Reducing social desirability: example crime

E.g. if want to find out how many people have committed a ***serious*** crime

→ Use graduation: softening up

Illegally
downloaded
music/TV?



Stolen from
a shop?



**Sold Class A
drugs?**

Or if you want to find out prevalence of a ***less serious*** crime

→ Use the reverse approach

Stolen a
vehicle?



Burgled a
house?



**Stolen from a
shop?**

Reducing social desirability: example voting

→ Normalising the behaviour

ESS R9 B13: Some people don't vote nowadays for one reason or another. Did you vote in the last [*country*] national election in [month/year]? Yes/No

Experiment by Belli et. al (2006) found that compared with a yes/no, voter turnout rates were reduced when these further options were added

1. I did not vote in the [Month and day] election.
2. I thought about voting this time but didn't.
3. I usually vote but didn't this time.
4. I am sure I voted in the [Month and day] election.

Reducing social desirability: example transgender

→ Change of mode and question formulation

Computer assisted personal interview (CAPI)	Self-completion
Q1: (DIRECT) How would you describe yourself... • as very prejudiced against people who are transgender, • a little prejudiced, • or, not prejudiced at all?	Do you think that prejudice against transgender people is always wrong, mostly wrong, sometimes wrong, rarely wrong or never wrong? PLEASE TICK ONE BOX ONLY (✓) Always wrong ☐ 1 Mostly wrong ☐ 2 Sometimes wrong ☐ 3 Rarely wrong ☐ 4 Never wrong ☐ 5 Can't choose ☐ 8

Reducing social desirability: example transgender

Table 3 Prejudice against people who are transgender

Would you describe yourself ...	%	
... as very prejudiced against people who are transgender	2	17%
... a little prejudiced	15	
... or not prejudiced at all	82	
<i>Unweighted base</i>		974
View of prejudice against transgender people	%	
Always wrong	53	38%
Mostly wrong	19	
Sometimes wrong	15	
Rarely wrong	3	
Never wrong	1	
<i>Unweighted base</i>		782

4. Sensitive topics

- Similar to social desirability bias as the questions are getting “personal”
 - Assure confidentiality
 - Explain why you are asking this question
 - Put your demographic questions at the end
 - Present sensitive questions toward the end (but not the very end)
 - Use ranges rather than asking for a specific numeric value
 - Ask a direct frequency question rather than using a yes/no filter, e.g.:

2019 Youth Risk Behaviors Survey:

45. During your life, how many times have you used marijuana?
- A. 0 times
 - B. 1 or 2 times
 - C. 3 to 9 times
 - D. 10 to 19 times
 - E. 20 to 39 times
 - F. 40 to 99 times
 - G. 100 or more times

Request for answer - Connected with research objectives

- Time reference
 - make it “the closer” to the present as possible
- Saliency or centrality
 - give more information
- Social desirability
 - alternatives like indirect question, images or letters (on showcards); self-completion mode; normalising the behaviour; graduation; highlighting the importance of the study and hence honesty; order and context of questions
- Sensitive topics
 - (previous slide)

Exercise

- Formulate a request for an answer to measure any of the following:
 1. Age of the respondent.
 2. If respondent has copied from someone in a test *or* academic work (plagiarism)
 3. *A sensitive topic of your choice*

Vaishali and Somaveh

- Formulate a request for an answer to measure any of the following:
 1. Age of the respondent
 - in which categorial age do you find yourself?
 - under 18
 - between 18 upto 35
 - 35-60...
 2. If respondent has copied from someone in a test *or* academic work (plagiarism)
 - have you ever thought copying from someone could be ethical?
 - i have thought about it but never did
 - I have never thought about it
 - I dont think its ethical at all
 - I thought its unethical but had to do it.
 3. *A sensitive topic of your choice*
 - *have you ever been harassed?*
 - *I have been forced to do sth*
 - *inappropriately toc*

Alisha, Lanre and Samad

- Formulate a request for an answer to measure any of the following:
 1. Age of the respondent
 - Which age group do you belong to?
 - 12-18
 - 19-24
 - 25-34
 - 34-64
 - 65 and older
 2. If respondent has copied from someone in a test *or* academic work (plagiarism)
 - Students often when under pressure copy their assignments or tests. Have you ever copied an assignment or paper from someone else or online sources in the past semester?
 - Never
 - Sometimes
 - Often
 - Always
 3. *A sensitive topic of your choice*
 - *Have you ever experience bullying in the past 2 weeks ?*

Jaiyi and Manfeng

- Formulate a request for an answer to measure any of the following:
 1. Age of the respondent. Please indicate your age in completed years: _____ (please write down the number)
 2. If respondent has copied from someone in a test *or* academic work (plagiarism)

Please indicate whether you have ever copied from someone in a test or academic work.

Answer options: Yes, No, Prefer not to answer

3. *A sensitive topic of your choice: Please indicate whether you have experienced financial difficulties during the last 12 months. Ans: Yes/No/Prefer not to answer*

Celal and Leonardo

- Formulate a request for an answer to measure any of the following:

1. Age of the respondent

What is your age group?

Please choose the category that applies to you.

- A. 18–24
- B. 25–34
- C. 35–44
- D. 45–54
- E. 55–64
- F. 65 or older
- G. Prefer not to say

REQUEST FOR AN ANSWER - WORDING

Language

- Should be specific, definitive, brief, simple and self-explanatory

To be avoided:

- No jargon or complex/technical language
- Not too simple
- Not too complex
- Words that mean the same to everyone

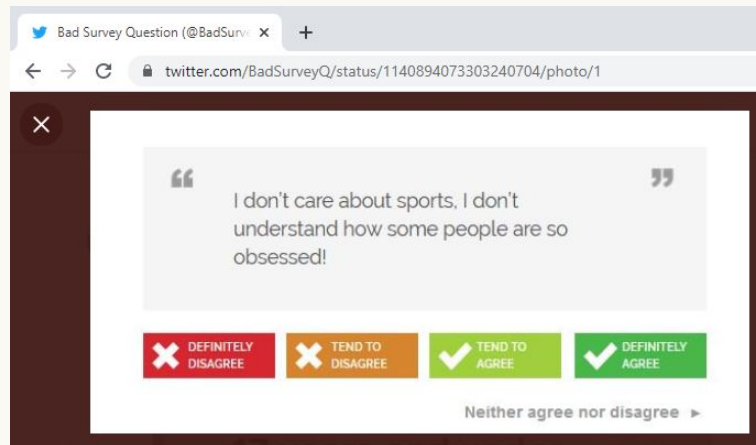
To be avoided: Double-Barreled Requests

- **Double-Barreled Requests** = asking for more than one concept.

Examples:

Do you agree with the statement that the asylum seekers should be allowed into our country [basic concept: right] but should adjust themselves to our culture [basic concept: norm]?

How well prepared do you think the health system and its medical staff was to cope with the COVID-19 pandemic and its consequences?



To be avoided: Loaded questions

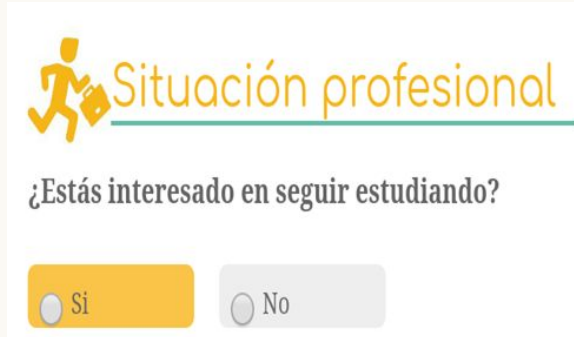
- **Loaded questions** = requests with implicit assumptions

Examples:

What is the best book you read last year?

Whom did you vote for?

Where do you like to exercise?



 **Situación profesional**

¿Estás interesado en seguir estudiando?

☒ Si ☐ No

Professional situation

Do you want to continue studying?

- Yes
- No

To be avoided: Leading questions

Leading questions = pushes for a certain response by making respondents feel that one kind of answer is more appropriate or 'normal' than others

Example:

How often do read?

Do you think society should do more for the elderly?

What makes our product better than others?

Do you prefer to be examined by a doctor of your own sex?

Better:

Have read books in the last year? If so, how many?

Would you rather be examined by a female or male doctor or does it not matter?

Example: leading question

4. Should Congress and the states eliminate so-called “gun-free zones” that leave innocent citizens defenseless against terrorists and violent criminals?

☐ Yes

☐ No

☐ No opinion

**Source: 2018 NRA National Gun Owner’s Action Survey*

To be avoided: negatively worded and double-negatives

Negatively worded questions

Example:

It is not important to me that my clothes are of the latest style



Double-negatives

Example:

I usually do not have one or more outfits that are not of the very latest fashion



OPEN AND CLOSED REQUESTS

Open or closed request

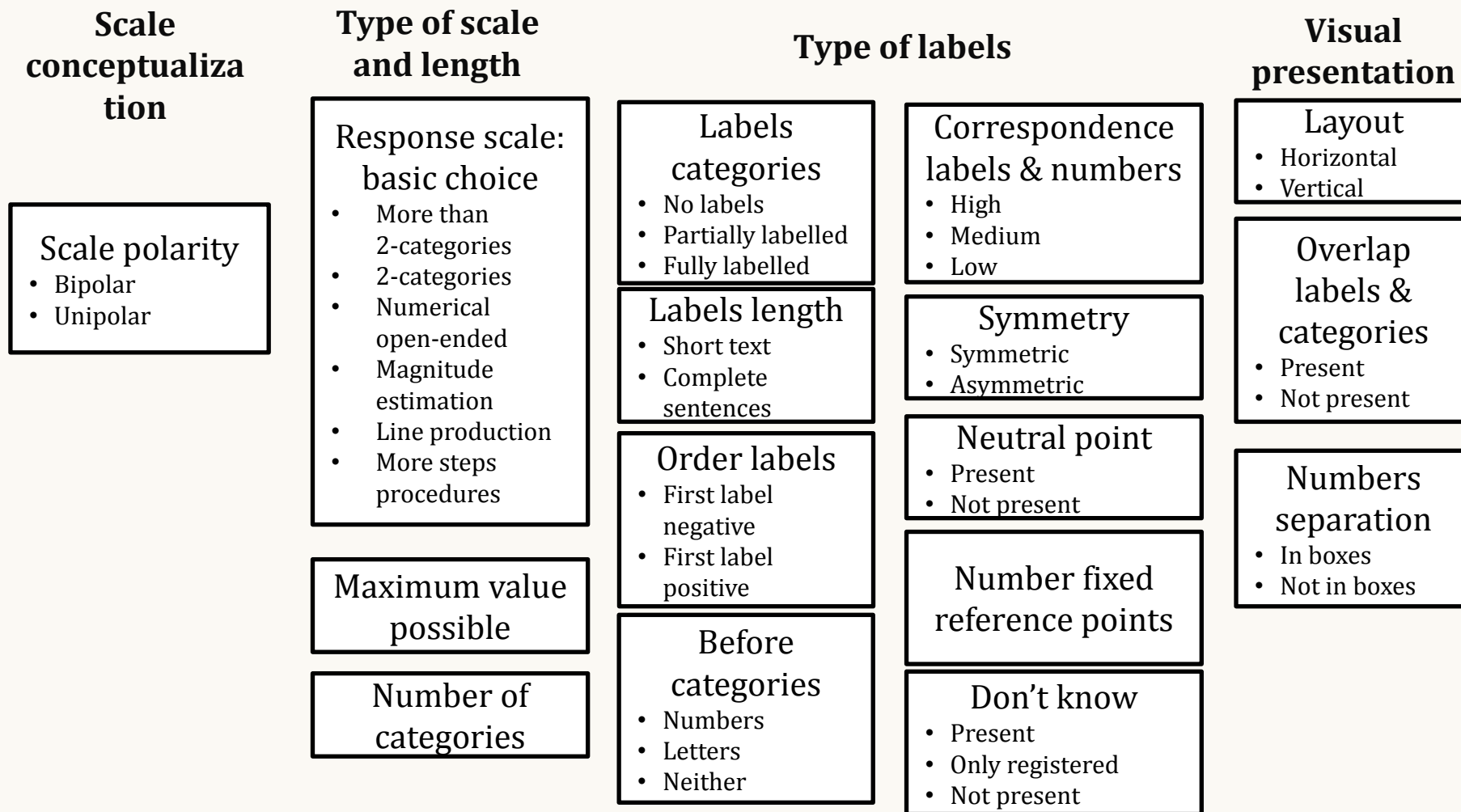
- **Open(-ended) requests** should provide other information than closed questions; otherwise closed questions are more efficient:
 - *What do you think about the presidency of X?*
 - *What brand of beer do you drink mostly?*
- These requests ask what is at the top of our mind. This is difficult to do with closed requests but requires more cognitive effort, longer completion time and coding afterwards.
- **Closed(-ended) requests** require response options from which the respondent chooses. Response categories can carry meaning and have implications for data analysis.

ANSWER CATEGORIES OR RESPONSE SCALES

Decomposition of a survey item

1. Introduction of the topic
2. Motivation for the respondent to answer
3. Information regarding the content
4. Information regarding a definition
5. Instruction for the respondent
6. Instruction for the interviewer
7. Request for an answer
8. Answer categories or response scales

Characteristics for answer options



Closed requests

Basic requirements any closed scale should meet:

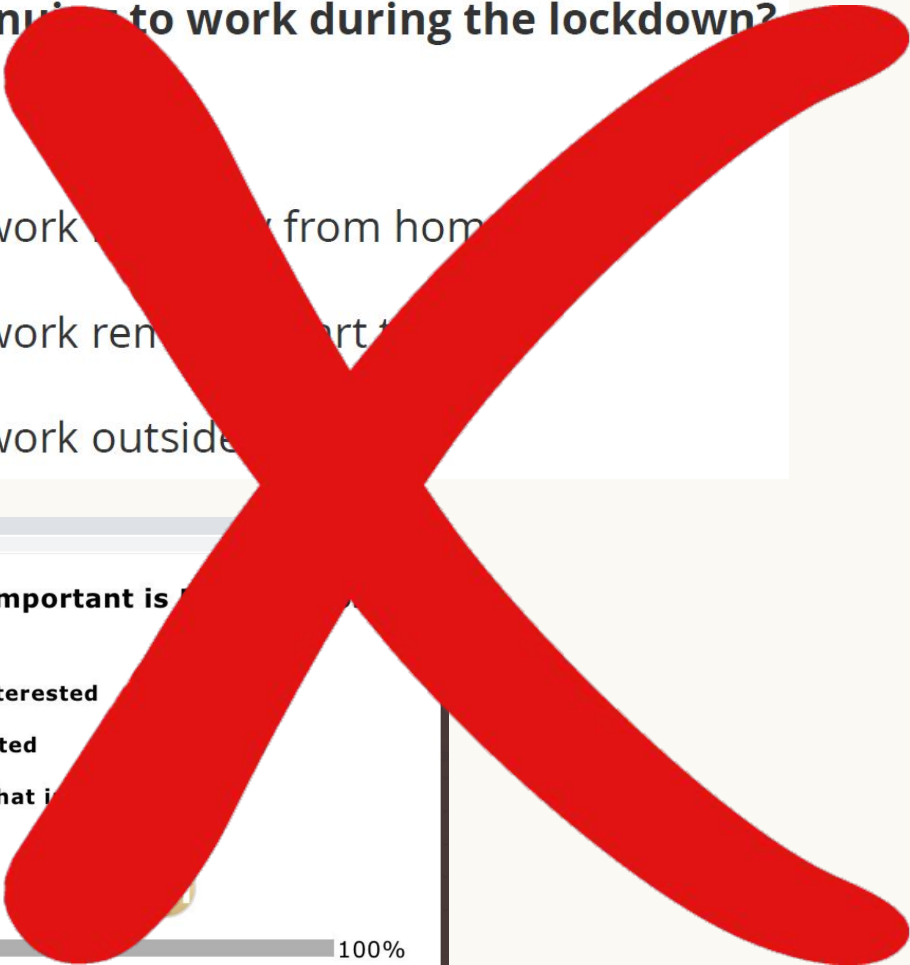
- Determined by the basic concept to be measured
- Has to be a *match* between request and responses
- The response alternatives should be *complete*
- They should be *exclusive* (if only one response allowed)
- All responses should refer to *the same concept*

Be careful!

- It seems obvious but in practice we see a lot of survey questions that do not follow these basic rules
- A few examples of what you should NOT do (even if others did it...)

Bad practise

- *Match* between request and responses



Are you continuing to work during the lockdown?

- ☐ No
- ☐ Yes, I work from home
- ☐ Yes, I work remotely part time
- ☐ Yes, I work outside

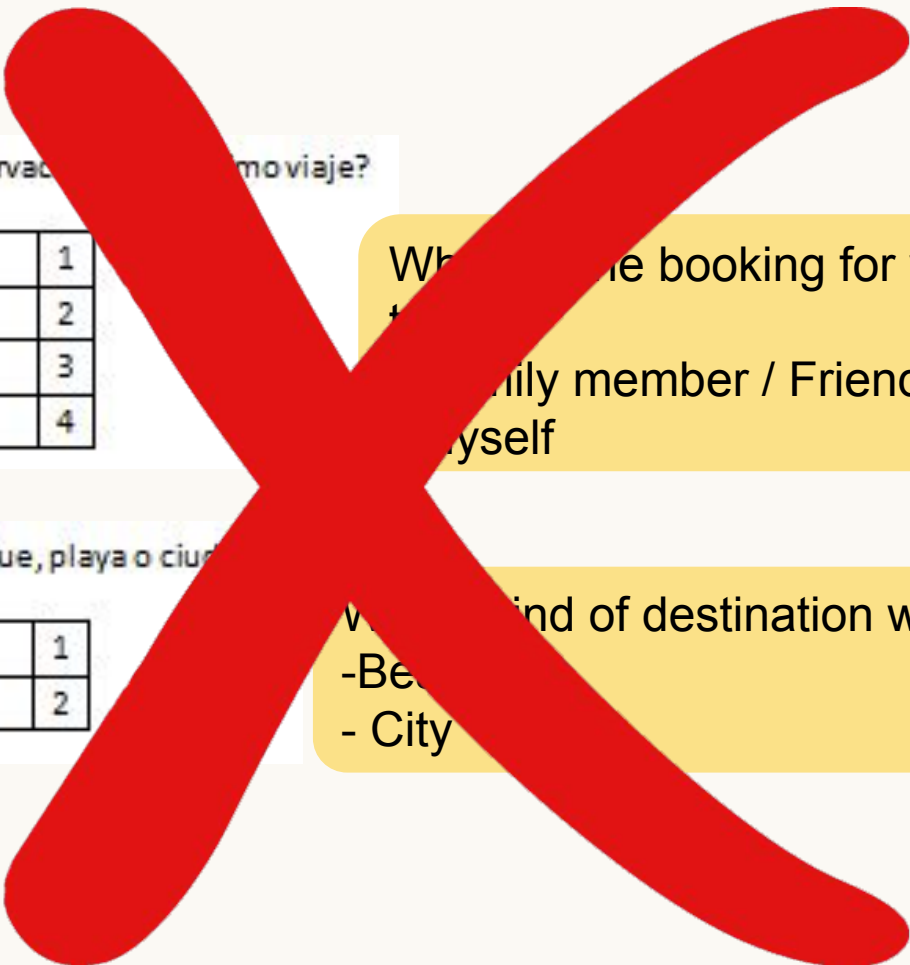
And how important is it for you?

- ☐ Very interested
- ☐ Interested
- ☐ Somewhat interested

0% 100%

Bad practise

- Response alternatives should be *complete* and *exclusive*



Q8 ¿Quién realizó la reserva de su último viaje?

Un familiar	1
Tus amigos	2
Mi pareja/ cónyuge	3
Yo mismo	4

When you were booking for your last trip, who did you book for?

- Family member / Friend / Partner

- Myself

Q10 ¿Qué tipo de destino fue, playa o ciudad?

Playa	1
Ciudad	2

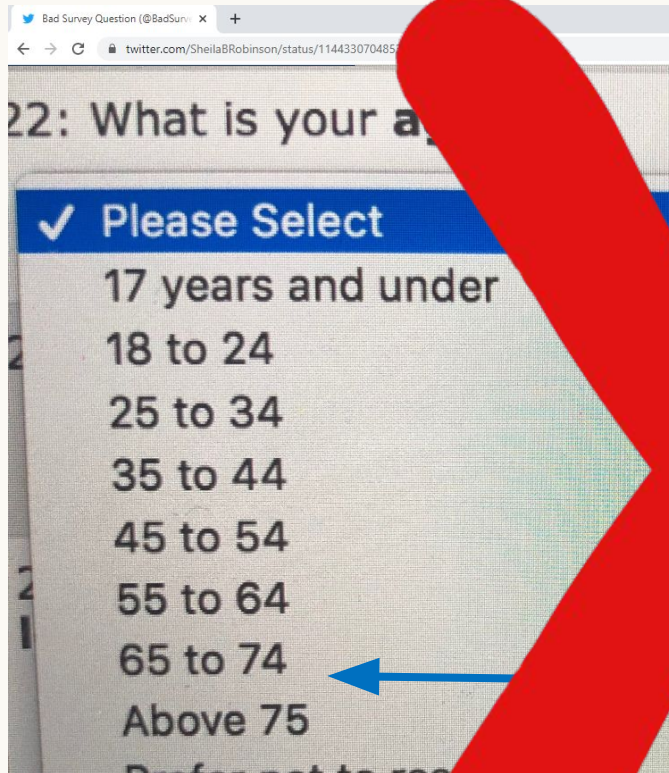
What kind of destination was it, beach or city?

- Beach

- City

Bad practise

→ What's wrong?



A screenshot of a survey question displayed on a mobile device. The question is "22: What is your age?". Below the question is a list of age ranges: "17 years and under", "18 to 24", "25 to 34", "35 to 44", "45 to 54", "55 to 64", "65 to 74", and "Above 75". A blue arrow points to the "65 to 74" option. A large red X is drawn over the entire survey question and its options.

22: What is your age?

✓ Please Select

- 17 years and under
- 18 to 24
- 25 to 34
- 35 to 44
- 45 to 54
- 55 to 64
- 65 to 74
- Above 75

Which option do you choose if you are 75 years old?

Bad practise

→ What's wrong?

learn about people's experiences
can continue to improve on the e
they fund in future. Do please take
moments to fill it out.

What is your age demographic?

- ☐ 18-27
- ☐ 28-37
- ☐ 48-57
- ☐ 58-67
- ☐ 68-77

<https://twitter.com/DocPhoenix/status/1398019190704021507/photo/1>

Your

18-22

☐ 23-30

☒ 30-40

☐ 40-50

☐ 50+

☐ Skip this question

<https://twitter.com/geraldldavidsson/status/1398019190704021507/photo/1>

5. How often did you go to the theatre on average in recent years?

- ☐ More than Twenty times a year
- ☒ Ten to Twenty times a year
- ☒ Four to Ten times a year
- ☐ Two to Three times a year
- ☐ Once a year
- ☐ Once every 2 years
- ☐ Once every 3 years

<https://twitter.com/geraldldavidsson/status/1398019190704021507/photo/1>

Bad practise

→ What's wrong?

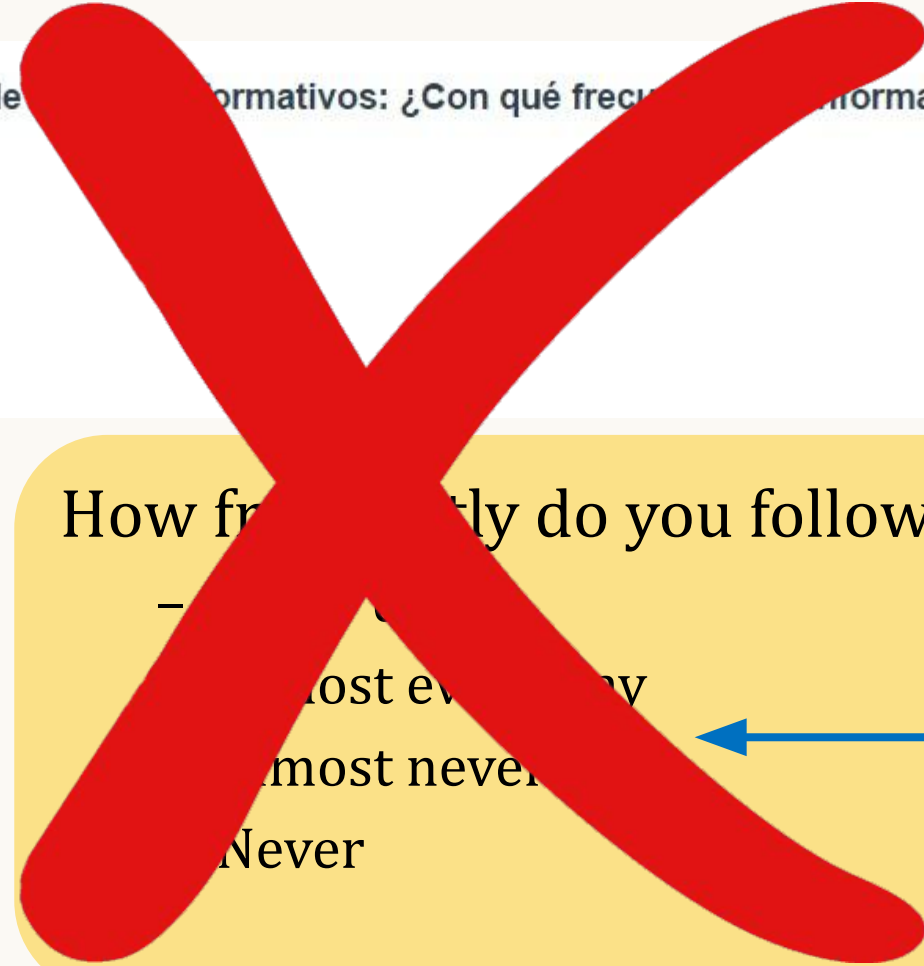
10. Ahora hablaremos de los medios informativos: ¿Con qué frecuencia recibes información sobre las noticias? *

- ☐ Cada día
- ☐ Casi cada día
- ☐ Casi nunca
- ☐ Nunca

How frequently do you follow the news?

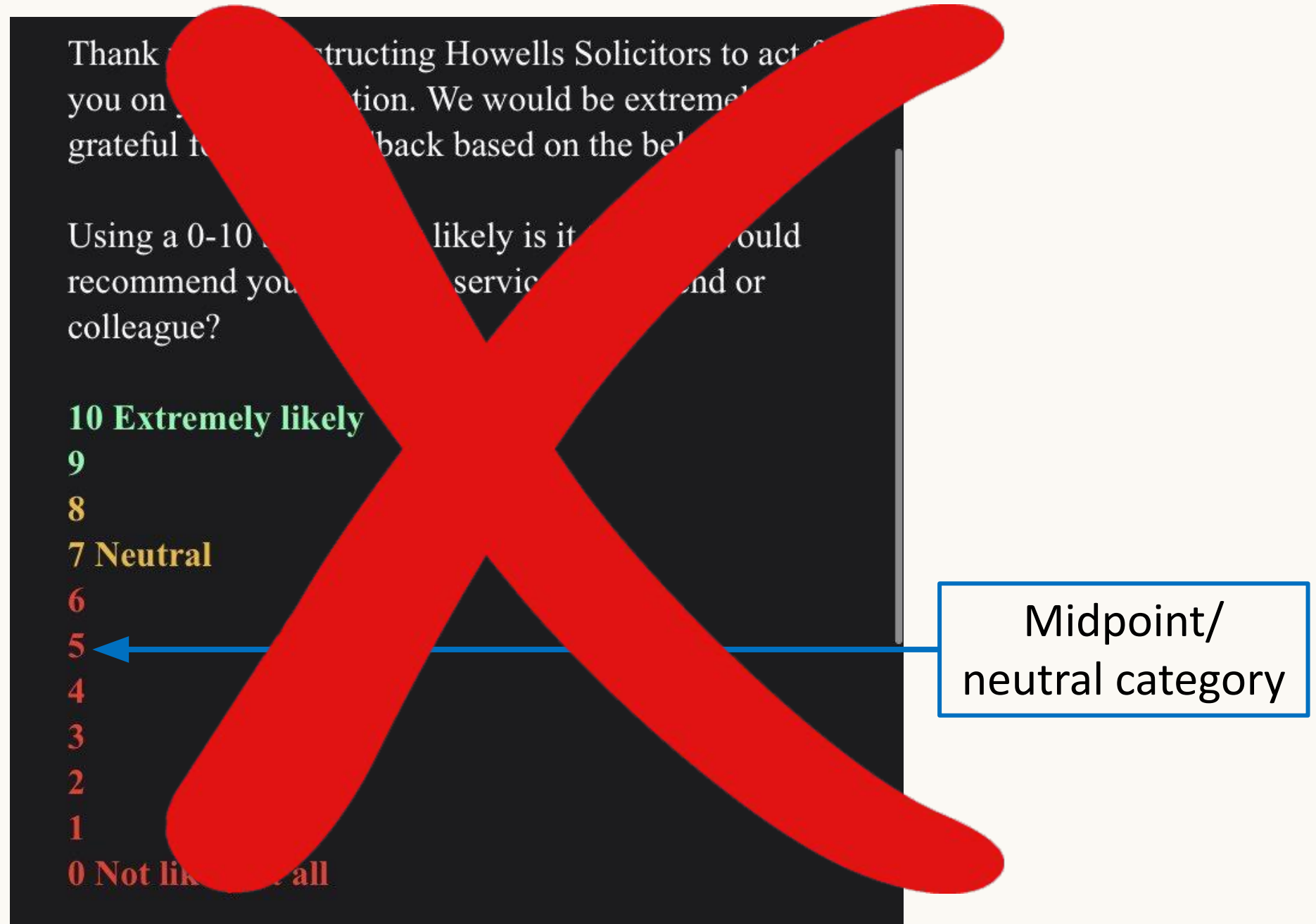
- ☐ Every day
- ☐ Most every day
- ☐ Sometimes
- ☐ Most never
- ☐ Never

Categories
missing



Bad practise

→ What's wrong?



Bad practise

→ What's wrong?

¿Tienes animales domésticos?

☐ Si, me encantan los perros

☐ No, prefiero los gatos

☐ Me gustan ambos

Do you have pets?

- I love dogs
- No, I prefer cats
- I like both

HOW WOULD YOU GRADE PRESIDENT TRUMP
IN THE NATIONS' FIGHT AGAINST THE VIRUS?

☐ SUPERB

☐ GREAT

☐ VERY GOOD

☐ NOT GOOD AT ALL

☐ NO OPINION

FOX BUSINESS
7:32P ET
LOU DOBBS TONIGHT | #Dobbs

Bad practise

→ What's wrong?

Are you interested in ... and
actively managing your ...
funds?

- ☐ Interested
- ☐ Not interested
- ☐ Very interested
- ☐ Slightly interested
- ☐ Prefer not to say

Order of response options

Bad practise examples on
Twitter: Bad Survey Question



<https://twitter.com/BadSurveyQ/status/1393233423045640195/photo/1>

Basic rules for closed requests

Basic requirements any closed scale should meet:

- Determined by the basic concept to be measured
- Has to be a *match* between request and responses
- The response alternatives should be *complete*
- They should be *exclusive* (if only one response allowed)
- All responses should refer to *the same concept*

→ **It is not enough to follow the basic rules**

Many other choices

- For a given question, many different scales can be used that all respect the basic rules
- Even if they respect the basic rules, some are much better than others!
- Difficulty is to evaluate the impact of all the different decisions we make when choosing one scale on the results and the quality of the data

Choices with respect to response options

Scale types:

Nominal (dichotomous)

Ordinal (vague quantifiers)

Continuous (line production and magnitude estimation)

Other choices to be made:

- Complete or partially labelling
- Number of categories
- Fixed reference points
- Symmetry
- Bipolar or unipolar
- Agreement between numbers and categories
- Middle category or not
- Don't know option or not
- Etc.

Response Scales types

Nominal:

1) unordered response categories

Example: *What was most boring about this course?*

1. *Lecture of the instructor*
2. *Sitting alone in front of the computer*
3. *Exercises*
4. *Other, please specify*

2) Dichotomous

Example: *Did you learn anything new in this course? 1.Yes 2.No*

Response Scales types

Ordinal scales = order of response categories (vague quantifiers)

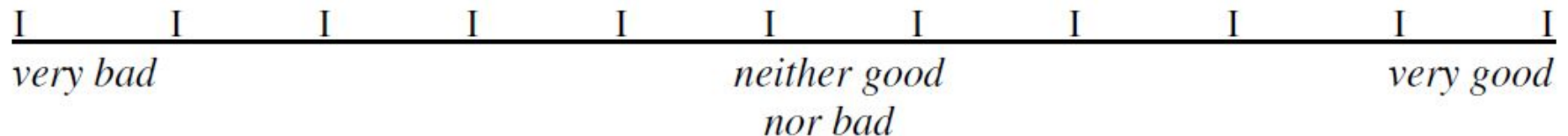
Example: *How good do you think Clinton was as president?*

1. *Very bad*
2. *Bad*
3. *Neither good nor bad*
4. *Good*
5. *Very good*

1. *Very bad*
2. *Rather bad*
3. *Bad*
4. *Neither good nor bad*
5. *Good*
6. *Rather good*
7. *Very good*

How good do you think Clinton was as president? Express your opinion in a number between 0 and 10 where 0 = very bad and 10 = very good.

How good do you think Clinton was as president? Express your opinion by placing an x at the point of the scale that best expresses your opinion.



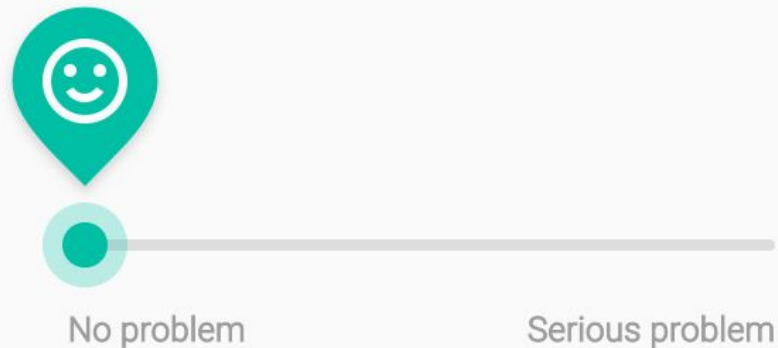
Response Scales types

Continuous: line production/sliders and magnitude estimation

- Examples:

How satisfied are you with your house? Express your opinion with a number between 0 and 100, where 0 is completely dissatisfied and 100 is completely satisfied.

And how serious a problem do you think climate change is at this moment?



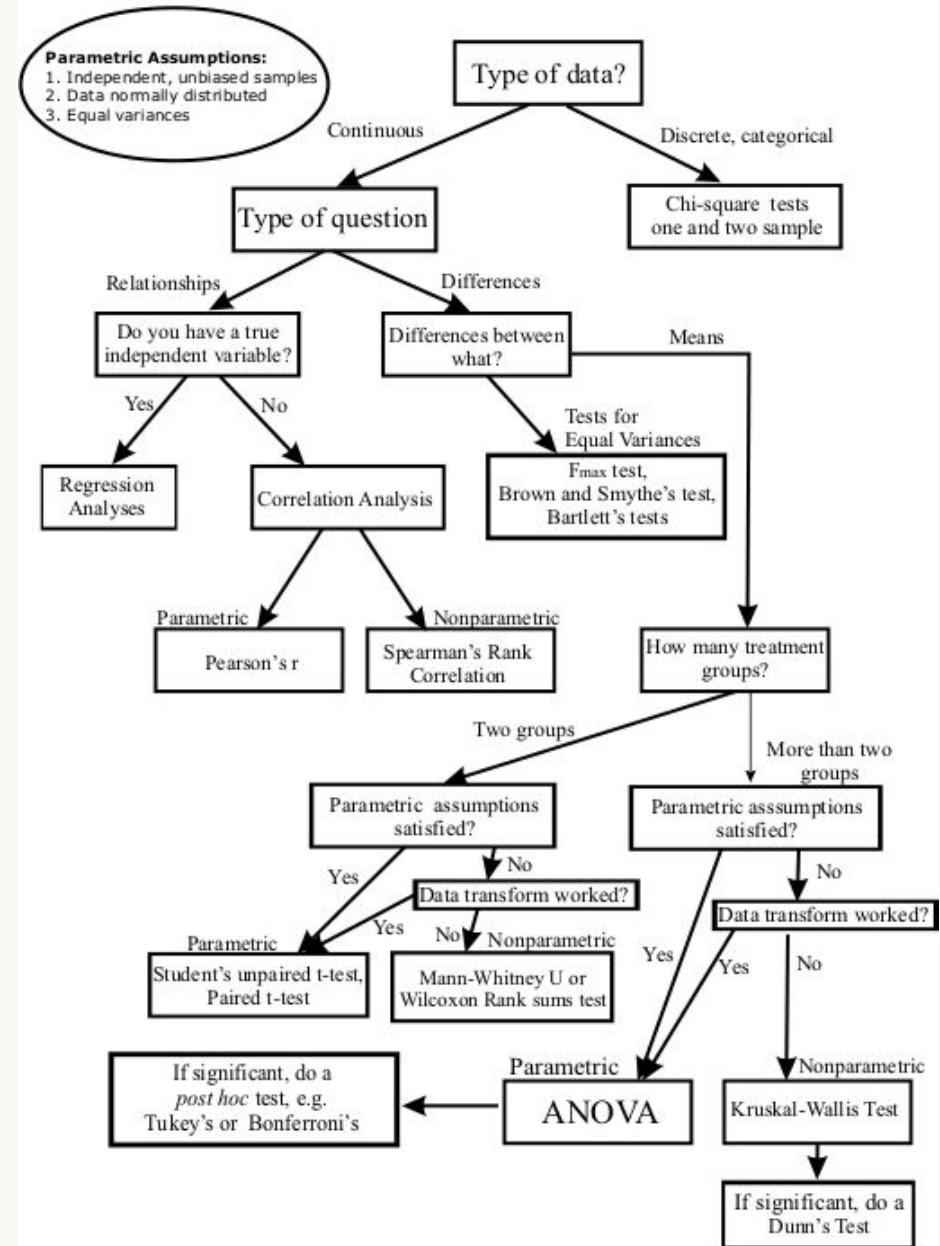
Scale types

Determined by research objective
and analyses plan!

UCLA Institute for Digital Research & Education Statistical Consulting							
HOME SOFTWARE RESOURCES SERVICES ABOUT US							
CHOOSING THE CORRECT STATISTICAL TEST IN SAS, STATA, SPSS AND R							
The following table shows general guidelines for choosing a statistical analysis. We emphasize that these are general guidelines and should not be construed as hard and fast rules. Usually your data could be analyzed in multiple ways, each of which could yield legitimate answers. The table below covers a number of common analyses and helps you choose among them based on the number of dependent variables (sometimes referred to as outcome variables), the nature of your independent variables (sometimes referred to as predictors). You also want to consider the nature of your dependent variable, namely whether it is an interval variable, ordinal or categorical variable, and whether it is normally distributed (see What is the difference between categorical, ordinal and interval variables? for more information on this). The table then shows one or more statistical tests commonly used given these types of variables (but not necessarily the only type of test that could be used) and links showing how to do such tests using SAS, Stata and SPSS.							
Number of Dependent Variables	Nature of Independent Variables	Nature of Dependent Variable(s)*	Test(s)	How to SAS	How to Stata	How to SPSS	How to R
1	0 IVs (1 population)	interval & normal	one-sample t-test	SAS	Stata	SPSS	R
		ordinal or interval	one-sample median	SAS	Stata	SPSS	R
		categorical (2 categories)	binomial test	SAS	Stata	SPSS	R
	1 IV with 2 levels (independent groups)	categorical	Chi-square goodness-of-fit	SAS	Stata	SPSS	R
		interval & normal	2 independent sample t-test	SAS	Stata	SPSS	R
		ordinal or interval	Wilcoxon-Mann Whitney test	SAS	Stata	SPSS	R
	1 IV with 2 or more levels (independent groups)	categorical	Chi-square test	SAS	Stata	SPSS	R
			Fisher's exact test	SAS	Stata	SPSS	R
		interval & normal	one-way ANOVA	SAS	Stata	SPSS	R

<https://stats.idre.ucla.edu/other/mult-pkg/whatstat/>

Flow Chart for Selecting Commonly Used Statistical Tests



<http://abacus.bates.edu/~ganderso/biology/resources/statistics.html>

Choices with respect to response options

Other choices to be made:

- Complete or partially labelling
- Number of categories
- Fixed reference points
- Symmetry
- Bipolar or unipolar
- Agreement between numbers and categories
- Middle category or not
- Don't know option or not
- Etc.

Use of labels

- **Label** = text that determine the meaning of a category
 - Can be used for
 - Each one of the response options = **completely labeled**
 - Some = **partially** labeled
 - None, i.e. only numbers or lines...
- **Always better to use labels at least for the end points**
- Using intermediate ones can be useful also but it depends on the number of response options and the kind of variable (numeric or not)

Use of labels – examples from ESS R8

The next set of questions are about yourself.

C7 How is your health³⁷ in general? Would you say it is... **READ OUT...**

...very good,
good,
fair,
bad,
or, very bad?

B37 CARD 14 Now thinking about the European Union, some say European unification²⁸ should go further. Others say it has already gone too far. Using this card, what number on the scale best describes your position?

Unification
has already
gone too far

Unification
should go
further

00 01 02 03 04 05 06 07 08 09 10

Number of categories

Theory of the information suggests that for bipolar scales:

- 2 points
 - information about the **direction** of the opinion
- 3 points
 - information about the direction + **middle category** (neutral)
- 4, 6, 8, 10,... points
 - information about direction + **intensity** of the opinion
- 5, 7, 9, 11,... points
 - information about direction + intensity + middle category

Number of categories

- More points = better?
 - More possibilities to nuance
 - More precise information
 - More similar to continuous variables→ So we can analyze the data as continuous which sometimes is easier
- In theory, yes, but in practice there is a limit!
 - Scale 0 to 100
 - Respondents tend to choose rounded values
 - Almost nobody chooses 42 or 81 ...

Number of categories

- Other problem for scales with many categories
 - “Error variance” increases with the number of categories
 - Some recommend to use 7 categories as most
 - no complete agreement, others say 5 or 11
- it depends on other characteristics of the scale

Number of categories

- To determine how many categories it is better to use, differentiate between **item specific (IS)** scales and **agree-disagree (AD)** scales
 - **AD: should not use more than 5 points** (Revilla, Saris & Krosnick, 2013)
 - **IS: might be that 11 works better**
- Take into account if the scale has **fixed-reference points** or not

AD versus IS

Cognitive steps needed for AD scales:

Respondents must read (or listen to) the statement and understand its literal meaning.

They must look deeper into the statement to discern the underlying dimension of interest to the researcher. (*comprehension*)

They must then place themselves on the dimension of interest. (*retrieval and judgement*)

They must translate this judgment into the agree/disagree response options (*response*)

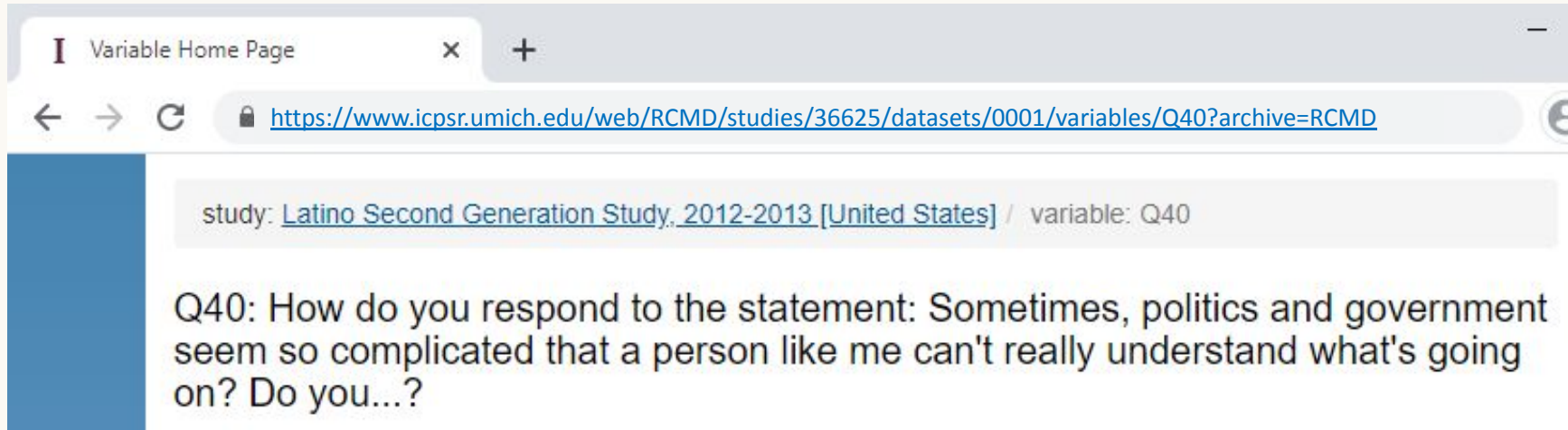
→ It is **simpler to skip this latter** step and simply ask respondents directly for their judgments.

Example of IS requests

- *Could you take a very active, quite active, limited role or no role at all in a group that is focused on political action?*
- *Can you understand and judge important political issues very well, well, neither good nor bad, bad, very bad?*
- *How often does it seem to you that politics and government are so complicated that a person like you cannot really understand what is going on: very often, quite often, sometimes, seldom, or never?*

→ In this case no translating to the AD scale is necessary. This is simpler for the respondents.

Example with AD scales



Agree strongly

Agree somewhat

Neither agree nor disagree

Disagree somewhat

Disagree strongly

Example with AD scales

How far do you agree or disagree that:

Sometimes politics and government seem so complicated that a person like me cannot really understand what is going on.

Instead of “sometimes”, we can use alternatives, for example:

1. *Very often politics and government seem so complicated that a person like me cannot really understand what is going on.*
2. *Quite often politics and government seem so complicated that a person like me cannot really understand what is going on.*
3. *Seldom politics and government seem so complicated that a person like me cannot really understand what is going on.*
4. *Never politics and government seem so complicated that a person like me cannot really understand what is going on.*

Another problem with AD scales

- The formulation of the statement is arbitrary
- None of these statements is better than the other. They have different positions on the response scale.
- But the choice of one or another has an impact on the results ☐
acquiescence bias
- Which one should we choose?

Which one should we choose ?

_____I_____

Only people close to the position of the item (I) will agree, all others disagree

_____I_____

People right of the item may disagree and at the left as well!

- People with very different positions can give the same answer
- These scales create problems with respect to the interpretation
 - **Much better to use IS scales!**
 - Saris et al. (2010) confirm by evaluating the quality using MTMM experiments that IS scales perform much better

Exercise

Do you strongly agree, agree, neither agree nor disagree, disagree or strongly disagree with the following statements:

- *“It is generally bad for our country's economy that people come to live here from other countries”*
- *“Cultural life is generally undermined by people coming to live here from other countries”*
- *“Our country is made a worse place to live by people coming to live there from other countries”*

→ Can you specify alternative IS request for this A/D battery?

ESS solution

[B41] Would you say it is generally bad or good for [country]’s economy that people come to live here from other countries? Please use this card.

Bad for the economy						Good for the economy				
0	1	2	3	4	5	6	7	8	9	10

[B42] And, using this card, would you say that [country]’s cultural life is generally undermined or enriched by people coming to live here from other countries?

Cultural life undermined						Cultural life enriched				
0	1	2	3	4	5	6	7	8	9	10

[B43] Is [country] made a worse or a better place to live by people coming to live here from other countries? Please use this card

Worse place to live						Better place to live				
0	1	2	3	4	5	6	7	8	9	10

Fixed reference points

Fixed reference point = category of the scale for which there is no doubt at all about the subjective scale in the respondent mind

For example, neutral point or category with a label that indicate in a way 100% clear for everybody that this category is the most extreme position possible

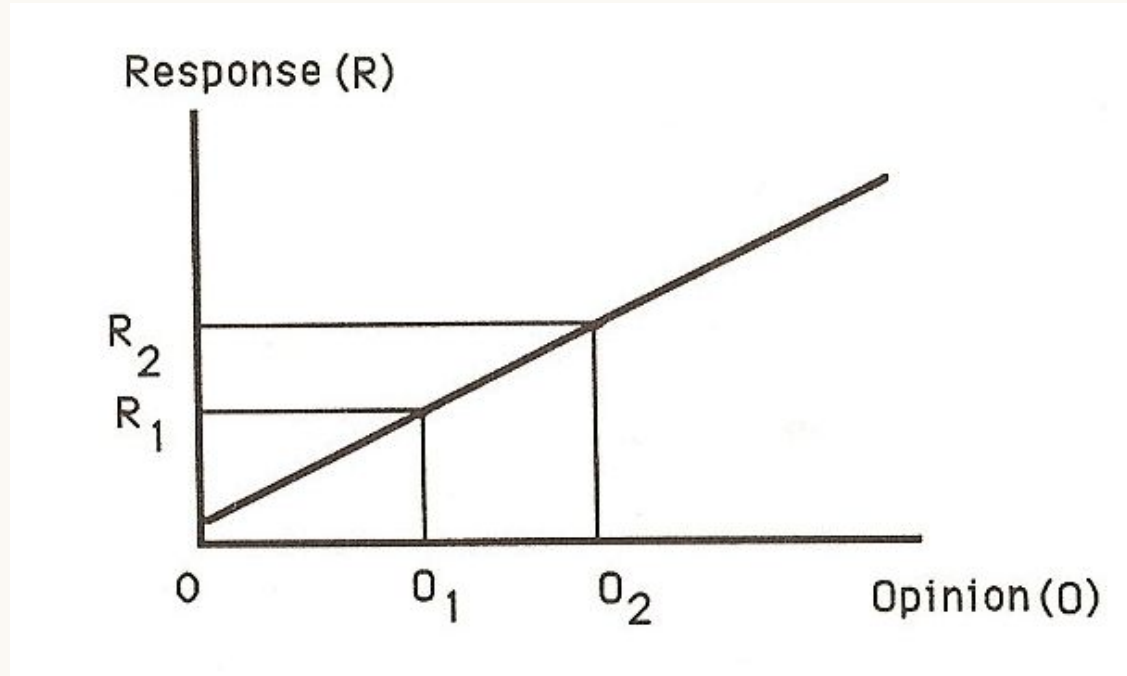
Examples:

“totally agree”, “complete trust”, “no trust at all”, ...

→ Why do fixed-reference points help?

Equality of the response function

- Usually it is assumed in survey analyses that the response functions are equal for all respondents
 - It means that **all respondents with the same opinion O_i will give the same answer R_i**



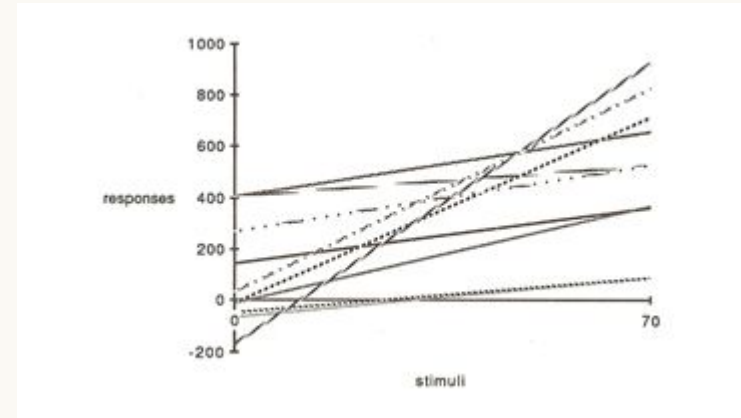
When is this true?

Without fixed reference points:

On this screen successive lines will appear
You are requested to express the length of each line in a number

If the line is long, type a large number
If the line is short, type a small number

Assign to the first line any arbitrary number.



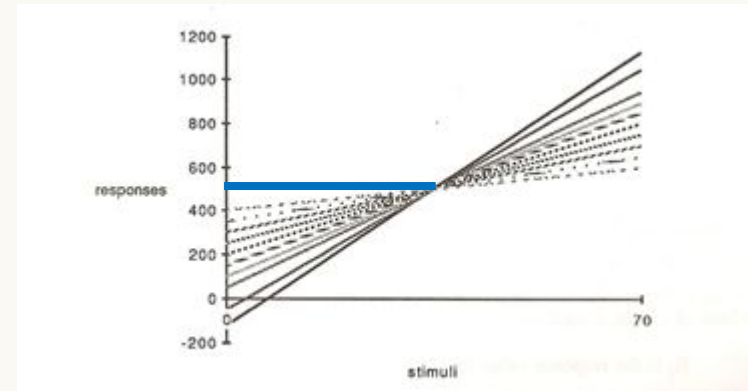
With **one** fixed reference points:

Below a line is shown

This line is called the standard line and is given the number 500.

Assign a number to the following line:

If you perceive this line as longer than the standard line, you give a larger number.
If you perceive this line as shorter than the standard line, you give a smaller number.



When is this true?

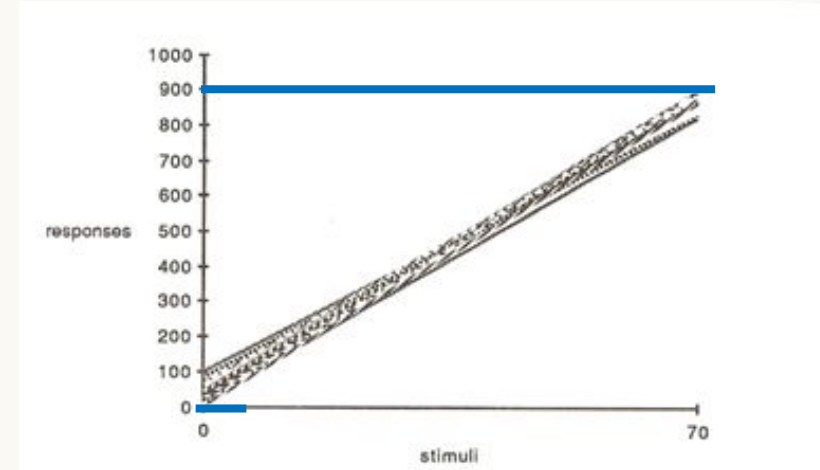
With **two** fixed reference points:

We will present to you several lines which vary in length.
The shortest line is of course a line with length zero.
The longest line is the line given below:

The shortest possible line is given the number 0
The longest occurring line is given the number 900

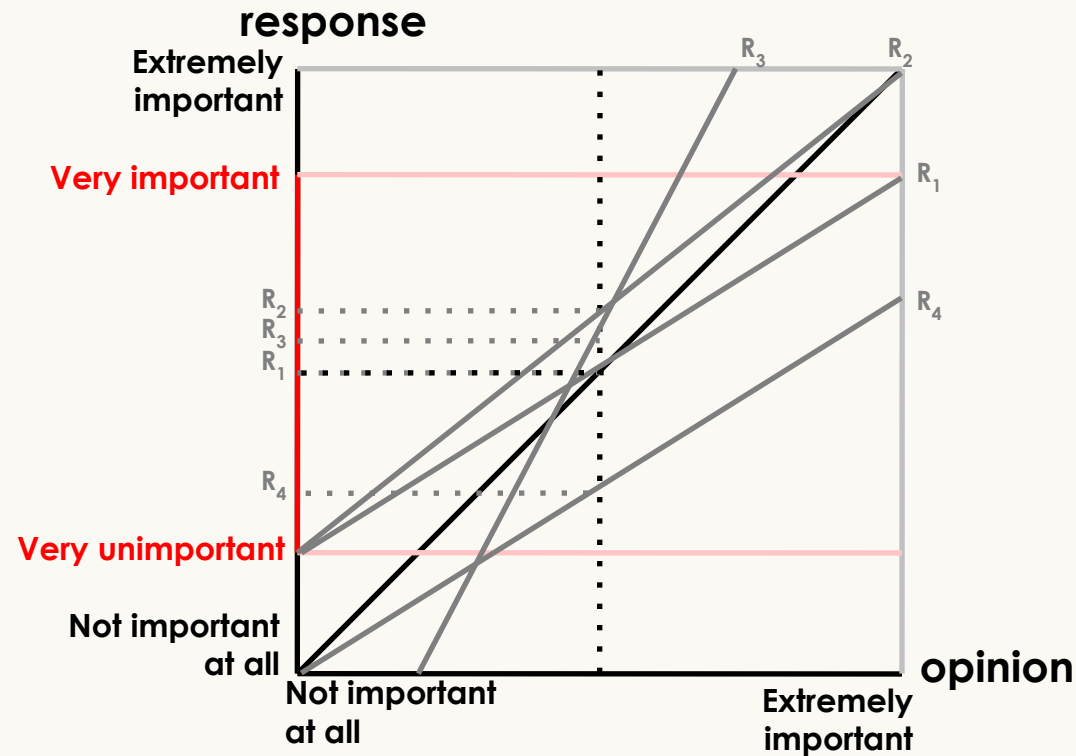
Assign a number to the following line

give a number between zero and 900



This is true when the 2 extreme points of the scale are fixed-reference points
→ thus we should at least have these 2 fixed reference points

Fixed reference points



→ without fixed references, answers are not comparable

Fixed reference points – ESS R9

Please tell me to what extent you think the governing parties in [country] are punished in elections when they have done a bad job?

The governing parties are not punished at all in elections when they have done a bad job

00 01 02 03 04 05 06 07 08 09 10

The governing parties are punished completely in elections when they have done a bad job

Please tell me to what extent you think the government in [country] protects citizens against poverty?

The government does not protect citizens against poverty at all

00 01 02 03 04 05 06 07 08 09 10

The government protects citizens against poverty completely

ESS R9 Multitrait-Multimethod (MTMM) experiment

	Method 1	Method 2	Method 3
Variations in characteristics	11-point	11-point	10-point
	'Apply scale'	'Item-specific scale'	'Item-specific scale'
	Battery of questions	Separate questions	Separate questions
	2 fixed reference points	2 fixed reference points	1 fixed reference point
	Implicit midpoint	Implicit midpoint	No midpoint

Method 1:

Does not apply at all

Applies completely

Governing parties in [country] are punished in elections¹⁷² when they have done a bad job.

0001020304050607080910

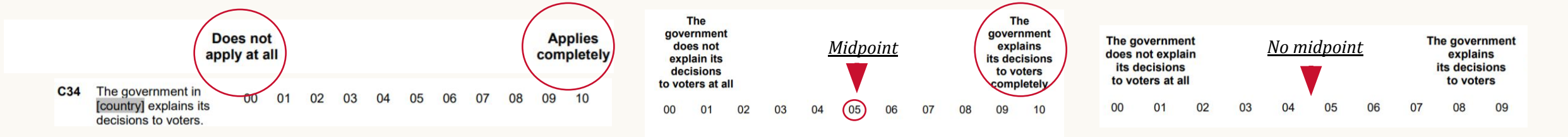
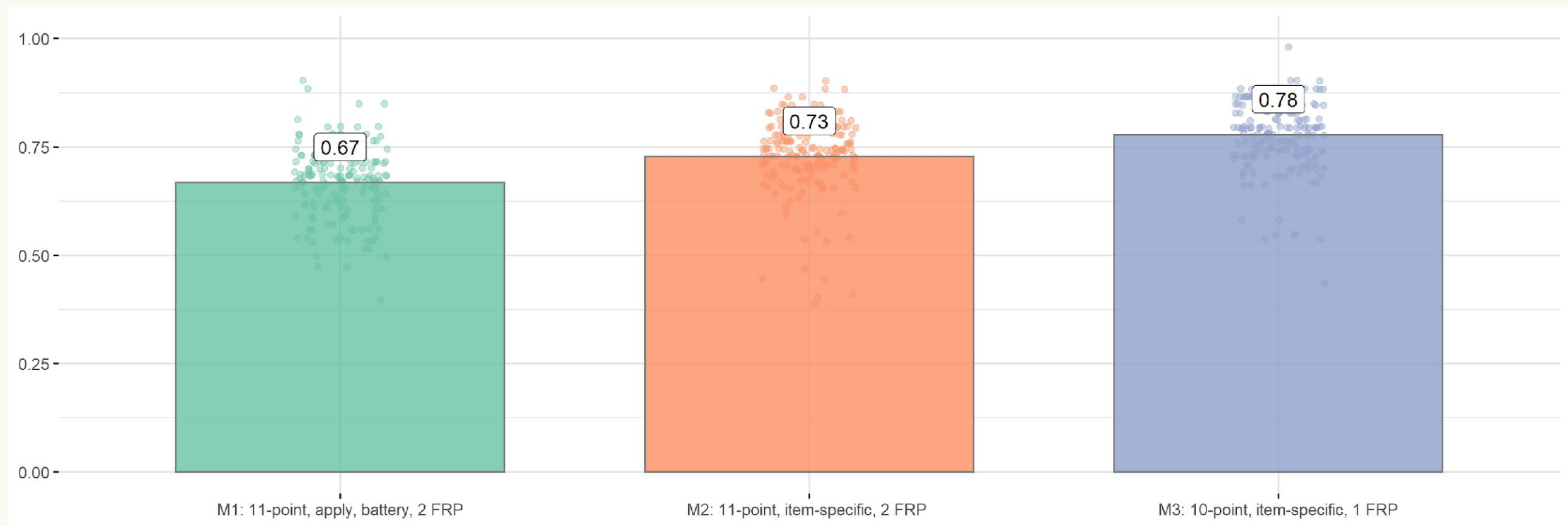
Method 3:

The governing parties are not punished at all in elections when they have done a bad job

The governing parties are punished in elections when they have done a bad job

00010203040506070809

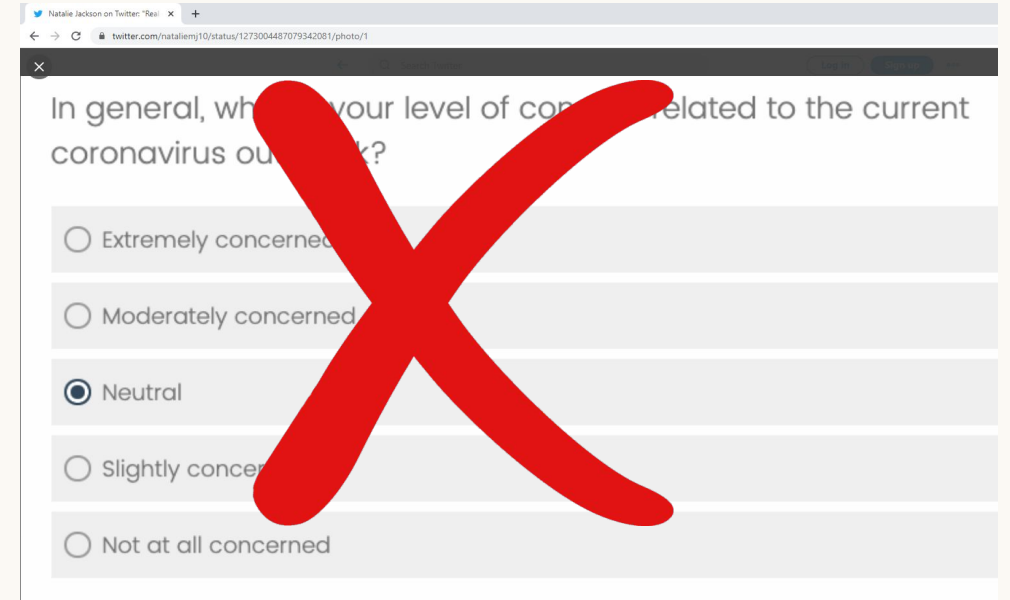
ESS R9 MTMM: Measurement Quality of response scales



→ fixed references only when topic allows for them

Unipolar and bipolar

- Unipolar
 - Never → always
 - Not at all satisfied → completely satisfied
- Bipolar:
 - Dissatisfied ↔ satisfied
 - Agree ↔ disagree



In general, what is your level of concern related to the current coronavirus outbreak?

- ☐ Extremely concerned
- ☐ Moderately concerned
- ☒ Neutral
- ☐ Slightly concerned
- ☐ Not at all concerned

- If the concept of interest is theoretically bipolar, better to use also a bipolar scale. If it is unipolar, better to use unipolar scale.
- Unipolar scales do not have a neutral midpoint!

Exercise

- We are interested in the following question and scale:
How satisfied are you with your life?
 1. *Not at all satisfied*
 2. *Satisfied*
 3. *Very satisfied*
- Is this scale:
 - Unipolar or bipolar?
 - Completely or partially labelled?
 - Does it have fixed reference points?
- Would you recommend to reformulate this question? If yes, why and how?

Symmetric

If the scale is bipolar and the categories on both sides reflect each other, e.g. completely agree, agree, NAND, disagree, completely disagree

→ **Recommended to use symmetric scales except if we know that the variable is really skewed**

1. *Very unhappy*
2. *Unhappy*
3. *Neither unhappy nor happy*
4. *Happy*
5. *Rather happy*
6. *Very happy*



1. *Not happy*
2. *A bit happy*
3. *Happy*
4. *Rather happy*
5. *Very happy*

Correspondence between numbers and labels

- if there is an order in the response categories (ordinal scale) and the number in front of the categories suggest the same order as the labels, then there is **high correspondence** between the labels and numbers
- If not = **low correspondence**

- Examples

Completely disagree	Disagree	Neither disagree nor agree	Agree	Completely agree	
-2	-1	0	1	2	High
5	4	3	2	1	Low

→ **Better to avoid low correspondence**

Don't know

- Can be mentioned explicitly
- Can be registered only (in f2f interviews)
- Can be not mentioned and not registered
- How to present it affects the results

→ In f2f interviews, 'registered only' often better

→ In web surveys: can use interactive features to do something in this direction

Example of problem with DK option

- 2 random samples of married women between 25-50 received the same question but half had an explicit DK option and the other half had not
- More than $\frac{1}{4}$ of the respondents chose DK when option explicit but give a substantive answer when it is only registered
 - (table: see next slide)

Example of problem with DK option

- Results

Filter		No filter	
The present government includes a secretary of state for women's emancipation. Do you think that this could improve the situation of women in society or not, or don't you have an opinion on that? (A: V129)		The present government includes a secretary of state for women's emancipation. Do you think that this could improve the situation of women in society or not? (B: V129)	
improve	54.7	72.3	improve
not improve	16.2	26.2	not improve
no opinion	29.1	1.6	(no opinion)
total		100.0 (179)	100.0 (191)
$\chi^2 = 55.69$; df = 2; p = 0.000			

Example of problem with DK option in web survey

Bad Survey Question (@BadSurvey) x

twitter.com/BadSurveyQ/status/1145652513036783617/photo/1

Log in Sign up

You might like

Emily Gilbert

Have you been on strike, been to a demonstration, signed a petition or conducted any other type of protest in the past year?

⚠ This question is mandatory. Please check at least one item.

☐ Strike

☐ Demonstration

☐ Petition drive

Next >

Did you know that a new Labour party leader was elected this week?

All adults (2763 GB adults - Apr 8, 2020)



YouGov | What the world thinks

yougov.com

<https://yougov.co.uk/topics/politics/survey-results/daily/2020/04/08/ad558/2>

→ In web surveys: better allow to skip than offer DK or force an answer

DK option in web survey

To what extent will scientific work priorities (especially those of political scientists) change in the years to come? Please indicate how much you agree or disagree with the following statements.

	Completely disagree	Somewhat disagree	Somewhat agree	Completely agree	Don't know	Prefer not to answer
The crisis has undermined the role of the social sciences.	☒	☐	☐	☐	☐	☐
We will all have to deal with policy-related questions in the years to come.	☐	☐	☐	☒	☐	☐
Fundraising and resource distribution method will dramatically change.	☐	☐	☒	☐	☐	☐
Experiments will be more important in all scientific contexts.	☐	☐	☒	☐	☐	☐
Pre-print and real-time results will become more important than time-consuming monographic works and peer-reviewed journals.	☐	☐	☐	☐	☐	☐

Only shows up after one items was not answered.

Neutral category

- Middle of the scale if symmetric bipolar scale with an odd number of categories (e.g. Completely disagree - Disagree - Neither disagree nor agree - Agree - Completely agree)
- The person has an opinion ($\neq$ “DK”) but this opinion is neutral (not positive nor negative)
- But sometimes respondents use it to avoid saying what they really think (“hidden DK”)

Neutral category

Examples: from [ESS R9 Source questionnaire](#):

Using this card, please say to what extent you agree or disagree with each of the following statements.

		Agree strongly	Agree	Neither agree nor disagree	Disagree	Disagree strongly
B33	The government should take measures to reduce differences in income levels.	1	2	3	4	5

B26 In politics people sometimes talk of “left” and “right”. Using this card, where would you place yourself on this scale, where 0 means the left and 10 means the right?

Left					Right					
00	01	02	03	04	05	06	07	08	09	10

Check-all-that-apply (CATA) or forced choice?

Forced-choice formats require more space and take more time than select-all-that-apply

Screenshots from Pew Research Center randomized experiment

The image shows two overlapping screenshots of survey questions. The top screenshot is a forced-choice question: "Which of the following, if any, have happened to you, personally?" with two options: "Yes, has happened to me" and "No, has not happened to me". The bottom screenshot is a CATA question: "Which of the following, if any, have happened to you, personally?" with a list of options: "Denied coverage by a health insurance company", "Over-charged by a mechanic or other service provider", "Were asked to pay a bribe", "Had your wallet or purse stolen", "Had a home break-in", "Had your identity stolen", and "None of these". The CATA question includes the instruction "[Check all that apply]" and allows for multiple selections.

Source: Survey of U.S. adults conducted July 30-Aug. 12, 2018.
"When Online Survey Respondents Only 'Select Some That Apply'"

PEW RESEARCH CENTER

Check-all-that-apply:

Which of the following extra-curricular activities would you be interested in participating in next year?

Please check all that apply.

- ☐ Art club
- ☐ Basketball
- ☐ Chorus
- ☐ Drama club
- ☐ ...
- ☐ Yoga
- ☐ Zoology club

Forced-choice:

Which of the following extra-curricular activities would you be interested in participating in next year?

Please check "yes" or "no" for each one.

- | Yes | No | |
|-----------------------|-----------------------|--------------|
| ☐ | ☐ | Art club |
| ☐ | ☐ | Basketball |
| ☐ | ☐ | Chorus |
| ☐ | ☐ | Drama club |
| ☐ | ☐ | ... |
| ☐ | ☐ | Yoga |
| ☐ | ☐ | Zoology club |

Gehlbach, H. (2015). Seven Survey Sins. *The Journal of Early Adolescence*, 35(5–6), 883–897. <https://doi.org/10.1177/0272431615578276>

→ **Forced choice preferable**

And there is much more...

- Presenting item-by-item or in a grid/battery?
- Horizontal or vertical orientation of the scale?
- Labels from positive to negative or the opposite?
- Radio buttons or dropdown menus, or sliders, or drag-and-drop...?
 - If dropdown menus: what visible first?
 - If sliders: where to place the marker at the beginning?
- Inclusion of images?
- Paging or scrolling?
- Etc.

Which characteristics impact the quality?

Scale conceptualization	Scale type and length	Scale labels		Scale visual presentation	
Scale evaluative dimension	Type of response scale	Verbal labels	Nonverbal labels	Type of visual response requirement	Overlap between labels
Scale polarity	Scale length	Verbal labels information	Order numerical labels	Slider marker position	Labels visual separation
Concept-Scale polarity agreement		Type of quantifier	Correspondence verbal-numerical labels	Illustrative scale format	Labels illustrative images
		Fixed reference points	Symmetry	Layout display	
		Order verbal labels	Middle alternative		
			Use of Don't know		

Impact on data quality
(based on literature review):

Found Not found Not clear

WHAT IS SQP?

SQP 3.0 is a survey quality prediction system for continuous questions used in survey research.

- ✓ CONSULT, COMPARE, AND EVALUATE CONTINUOUS QUESTIONS
- ✓ DESIGN NEW QUESTIONNAIRES
- ✓ CORRECT FOR MEASUREMENT ERRORS

WHAT CAN YOU DO WITH SQP?



Consult

SQP question database and
quality predictions



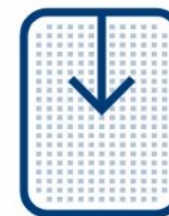
Add & contribute

Add a new question and
obtain its quality prediction



Compare

Compare questions and
their quality predictions



Download

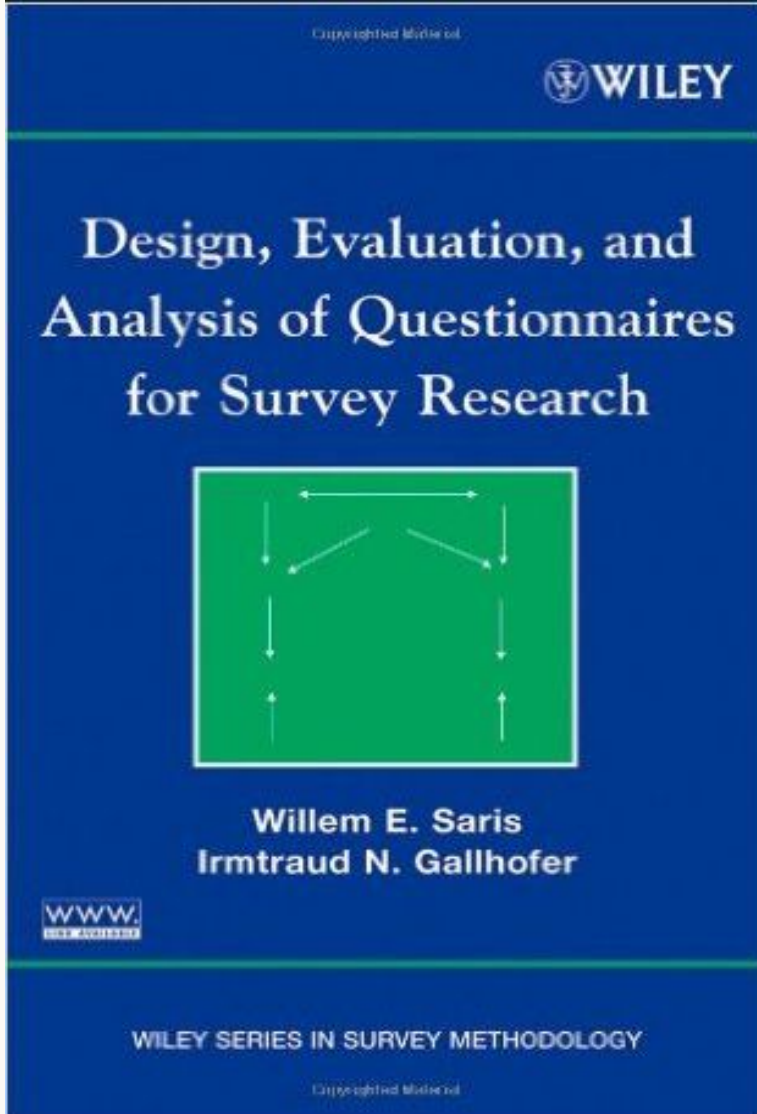
Download any information
from the database

Summary

Many decisions have to be made to design a survey question

- We have given some basic rules that should always be respected
- We have proposed some examples of scales characteristics that seem to be preferable to others based on empirical evidence and/or theories
- But the final impact on the quality depends on the specific question and on the interaction between the different decisions
- So in each particular case, we have to think deeply about each of the decisions we are making and justify why we use one scale and not another
- It is very difficult because of the interactions to know what to choose
- A possible solution: use the SQP software

Main references



Revilla, M., D. Zavala-Rojas, and W. Saris (2016). “Creating a good question: How to use cumulative experience”. In C. Wolf, D. Joye, T.W. Smith and Y.-C. Fu (editors), *The SAGE-Handbook of Survey Methodology*. SAGE. ISBN: 9781446282663. Chapter 17, pp.236-254.

Weber, W., I. Gallhofer, and W. Saris (2020) “Design of Survey Questions” SAGE Research Methods Foundations, 2020. SAGE Research Methods.